

AI Systems: Three Lines, One Map

Frontier AI Roadmap

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How to use this book

Where vision, language, and action meet.

A personal, worked-through companion to the **Learning Roadmap · System Map Edition** — a parallel curriculum for modern LLM architectures, computer vision on video, and deep reinforcement learning, with every point where the fields genuinely share an idea called out explicitly.

This is a study-notes book, not a survey. I wrote it the way I actually learn: the intuition first, the mechanism second, and the honest "here's what you should sit down and *do*" last. It assumes you'll work through it — implement a thing, read a paper, question a claim — not just read it end to end.

The map

There are **three lines**, and they are not the same length. Each runs as long as its subject actually requires:

- **Part 0 — Prerequisites** (6 stages): the programming, math, and core-ML vocabulary every later stage quietly assumes. Start here if you have no ML background.
- **Part I — Modern LLM Architectures** (8 stages): the **language line** — from recurrence to agents that plan and call tools.
- **Part II — Computer Vision on Video** (10 stages): the **vision line** — from the CNN to perceiving motion, tracking people, and generating video outright. The longest line, because it has more independently-teachable ideas.
- **Part III — Deep Reinforcement Learning** (7 stages): the **action line** — from the formal decision-making framework to agents that plan inside learned world models.
- **Part V — The Terminus** (1 stage): where all three lines end at the same station — generalist multimodal agents that perceive, reason in language, and act.

Interchange stations

The whole reason to study three fields as one map is that they don't just run in parallel — they **converge**. Where two lines meet, I've flagged it as an interchange, and the same idea gets described from both sides.

Stage	Interchange	The shared idea
LLM 1 / CV 7	LLM ↔ Video CV	Recurrent sequence models — the machinery language built on before attention, still the live temporal model in action recognition.
LLM 3 / CV 2	LLM ↔ Video CV	The transformer block is the same architecture — one reads tokens, the other reads image patches.

Stage	Interchange	The shared idea
CV 5 / RL 4	Video CV $\leftrightarrow$ Deep RL	A world model that predicts future frames is the same object as a learned environment model an RL agent plans inside.
LLM 6 / RL 5	LLM $\leftrightarrow$ Deep RL	RLHF and DPO — turning human preference into a training signal — is literally the same technique.
LLM 7 / CV 8	LLM $\leftrightarrow$ Video CV	Vision-language fusion, seen from the generative side (LLMs) and the embedding side (video-language models).
Terminus	All three	Vision-language-action models: one system, three fields fused.

The arc — why this order

Each part hands off to the next. **Part 0** ends on building a neural network by hand. **Part I** begins by asking how to make sequence models stop reading one word at a time. **Part I** ends on agents that loop over actions; **Parts II and III** pick up where perception and action become the point. **Part II** ends on generative video $\approx$ world-model prediction; **Part III** picks up planning inside world models. **Part III** ends on LLM-guided agents; **Part V** fuses everything into one system.

Read it as one journey, not five bucket-lists.

How to use this book

- **It's a map, not a syllabus — so start wherever you like.** You don't have to read it front to back. If a concept interests you, look it up in the **concept index** at the back of the book; it tells you where it's covered and the **path** to get there. Every chapter opens with a **Getting here** line for exactly this reason, so any stage can be an entry point.
- **Already know the foundation?** Skip Part 0 and start at Part I. Each chapter is self-contained enough that you can dip in, but the order is deliberate — later chapters build on earlier ones.
- **Each stage chapter** has the same shape: a one-line premise, the fleshed-out idea, why it matters, what to actually learn (with *how* to practice), the references, and a **Reflection + Try this week** to make it stick.
- **Interchange chapters** cross-reference their partner so you can read the same idea from both entry points — the lateral shortcuts between the three lines.
- **The terminus** (Part V) is the payoff. Every earlier chapter is, quite literally, a component a vision-language-action model is assembled from.

Everything here is my working understanding, grounded in real sources (listed per chapter) and checked against the 2024–2026 landscape so it isn't stale. Where a claim is contested or only shown in one paper, I say so.

Part I

Part 0 — Prerequisites

Contents of this part

- CH01 · Programming for ML
- CH02 · Linear Algebra Essentials
- CH03 · Calculus & Optimization
- CH04 · Probability & Statistics
- CH05 · Core Machine Learning Concepts
- CH06 · Neural Networks (MLPs, Forward & Backward)

Start here if you're new. The programming, math, and core machine-learning ideas every later stage in this map quietly assumes you already have.

If any of these six feel solid already, skip ahead — the chapters are written so you can jump straight into Part I. But don't skip them on a guess. The point of this part isn't to be thorough for its own sake; it's to make sure that when you meet attention, backpropagation, and policy gradients later, you're wrestling with the *idea*, not fighting syntax or notation you never quite learned.

There's a deliberate arc here, and it's tighter than it looks:

1. **Programming for ML** — the tool of the trade. Python plus the array library (NumPy) that every other numerical library speaks, and PyTorch, where the actual deep learning happens.
2. **Linear Algebra Essentials** — the *notation* every neural network is written in. A forward pass is, mechanically, matrix multiplications.
3. **Calculus & Optimization** — the *mechanism* of learning: gradients, and walking downhill.
4. **Probability & Statistics** — the *language* every model's uncertainty and every training objective is written in.
5. **Core ML Concepts** — the *vocabulary* of "training a model" itself, before any of it involves a transformer.
6. **Neural Networks** — where the pieces become a model: layers, the forward pass, backpropagation, and the training loop.

Programming → notation → mechanism → language → the whole idea → the network itself. Each one is the foundation the next builds on, and all six together give you the shared baseline that Parts I, II, and III use constantly without re-explaining.

The single highest-value move across this whole part is **one implementation**: take something you already understand mathematically — linear regression via gradient descent is the classic — and build it from raw NumPy, no framework doing the calculus for you. It's tedious exactly once, and it turns every later mention of "gradients," "loss," and "optimization" in this map from abstraction into something you've already done by hand.

Let's start with the tool.

Chapter 1

CH01 • Programming for ML

The practical baseline: Python, and the libraries everything else in this map is built on.

1.1 Getting here

Path: start here — no prerequisites.

1.2 Premise

Every stage in this book assumes you can read and write working Python, comfortably enough that **code stops being the obstacle** and the underlying idea becomes the thing you're actually wrestling with.

1.3 The bar is "comfortable," not "engineer"

Let's be honest about the target, because it shapes how you should spend your time. You do **not** need to be a software engineer. You don't need to know software design patterns, package publishing, or profiler internals. What you need is to not be fighting syntax while you're trying to understand backpropagation.

Concretely, that means the core language — functions, classes, loops, list/dict comprehensions — plus enough comfort with the command line to install packages, run a script, and read a stack trace without panic. A stack trace isn't a judgment; it's a map. The last few lines tell you the file and line where things went wrong, and the type of error tells you what category of mistake it is (`TypeError` is almost always a shape or type mismatch — the single most common bug in this entire field).

1.4 The three libraries that do almost all the work

On top of the language, three libraries carry essentially all the practical load in this field, and you should get genuinely fluent in the first two:

- **NumPy** — fast array operations. This is the vocabulary every other numerical library speaks. The mental shift that matters most is moving from *element-by-element* loops to *whole-array* thinking: `a * b` multiplies two arrays elementwise, `a @ b` does a matrix product, and broadcasting lets you combine arrays of different shapes without writing a loop. Vectorization (doing the operation on the whole array at once) isn't just faster — it's the way every deeper library is written, so thinking in it transfers directly.
- **PyTorch** — the dominant deep learning framework (as of this writing; more on the framework question in a moment). It gives you **tensors** (NumPy arrays with GPU support and a few

extras), **autograd** (automatic differentiation — it computes gradients for you), and the building blocks for neural networks (**nn.Module**, layers, optimizers). NumPy fluency transfers almost one-to-one: a tensor *is* an array with training superpowers.

- **Pandas** — tabular data handling. It shows up constantly in any stage involving real datasets rather than clean, prepackaged benchmarks. Worth knowing, less critical than NumPy.

1.5 The one exercise that's worth more than the rest

The single highest-value thing you can do at this stage is **implement something you already understand mathematically, from raw NumPy, with no framework doing the calculus for you**. Linear regression via gradient descent is the canonical choice:

1. Write the prediction as `y_hat = X @ w + b`.
2. Write the loss as mean squared error.
3. Derive (or look up) the gradient of the loss with respect to `w` and `b`.
4. Write the update loop by hand — `w -= lr * grad_w` — for a few hundred steps.
5. Watch the loss fall, then verify your `w` matches a closed-form or **sklearn** answer.

It's tedious exactly once. And it's worth it, because it makes every later mention of "gradients" and "optimization" in this document concrete rather than abstract. When Part III talks about a policy gradient being "an expectation over samples," you'll recognize the shape of it from having hand-rolled a gradient loop.

1.6 A note on the framework question (2024–2026)

The roadmap was written in the PyTorch-everywhere era, and for most learning purposes that's still the right call: PyTorch has the largest ecosystem, the best tutorials, and it's what most papers and courses assume. But it's worth knowing the landscape moved. **JAX** (with its functional, `jit`-friendly style) became increasingly prominent in research through the early-to-mid 2020s, and a fair number of frontier labs build on it. **PaddlePaddle** is significant in the Chinese ecosystem, and **MLX** is Apple's framework for Apple Silicon.

My advice: **learn PyTorch**. It's the lowest-friction path into every later chapter, and the core concepts (tensors, autograd, modules) transfer to JAX or anything else almost unchanged. Once you can read and modify PyTorch code fluently, picking up JAX is a days-long exercise, not a re-education.

1.7 What to actually learn

- **Core Python**: functions, classes, comprehensions, basic file I/O.
- **NumPy**: array creation, indexing, broadcasting, vectorized operations — and the habit of thinking in whole-array operations.
- **PyTorch tensors and autograd**, at the level of running and modifying existing code (you'll build your own models from scratch starting in Part 0, Stage 5).
- **Reading and debugging a stack trace** without dread — it's a map, not a report card.

1.8 References

- The Python Tutorial — official docs.
- NumPy Quickstart.
- PyTorch: Learn the Basics.

1.9 Reflection

What's the last time a stack trace (or error message) actually *told* you where to look, versus when it froze you? What did the error *type* reveal?

1.10 Try this week

Implement linear regression via gradient descent from raw NumPy — no PyTorch, no `sklearn`. Get the loss to fall, and confirm your learned weights match a library's answer within a small tolerance. That's the whole exercise. Then, for bonus points, repeat it in PyTorch and notice what autograd removes from your life.

Chapter 2

CH02 • Linear Algebra Essentials

Vectors, matrices, and the operations every neural network is built from.

2.1 Getting here

Path: start here (or right after Programming for ML).

2.2 Premise

A neural network's forward pass is, mechanically, a sequence of matrix multiplications and simple elementwise nonlinearities. Linear algebra isn't background theory here — it's the **literal notation** everything else in this map is written in.

2.3 Reframe: matrices are transformations, not boxes of numbers

The fastest way to stop fearing linear algebra is to stop reading it as arithmetic and start reading it as *geometry*. A **vector** is an ordered list of numbers, and it's useful to hold both pictures in your head at once: it's a **point** in space, and it's a **direction with a magnitude**. A **matrix** is a *linear transformation* — something that takes vectors in and produces vectors out, stretching, rotating, or projecting the space they live in. **Matrix multiplication is function composition**: applying one transformation after another.

Once that framing clicks, phrases that otherwise read as mystifying jargon become literal, visualizable matrix multiplies. The roadmap throws this one at you constantly across Parts I and II: "*project the patch embeddings into the language model's embedding space.*" That's not magic — it's one matrix multiply that rotates/compresses a vector from one coordinate system (vision) into another (language). You've seen it; it's just a transformation.

2.4 The operations worth real fluency (not just recognition)

A handful of operations recur so constantly that you should aim for fluency, not just "oh I've seen that":

- **The dot product** measures how *aligned* two vectors are — how much they point the same way. This is not a curiosity: it is the literal operation inside self-attention's query-key comparison (Part I, Stage 2). When I tell you attention "measures how much each token should attend to another," the mechanism is a dot product. Understand it geometrically (projection, cosine of the angle) and a huge chunk of the transformer demystifies itself.

- **Matrix-vector and matrix-matrix multiplication** — including getting genuinely comfortable with *shapes*. Shape mismatches are the single most common bug in this field, and they're usually a one-line fix if you can look at $\mathbf{X} @ \mathbf{W}$ and know what shape the output *must* be before running it. Learn the rule of thumb: an $(\mathbf{m} \times \mathbf{n})$ and an $(\mathbf{n} \times \mathbf{p})$ produce an $(\mathbf{m} \times \mathbf{p})$; the inner dimensions have to match.
- **Eigenvalues/eigenvectors** — the "natural directions" of a transformation: the directions that, when transformed, just get scaled rather than rotated. They show up whenever a method involves finding a matrix's stable directions — PCA-style dimensionality reduction is the recurring one here, and they reappear in dynamics analysis. You need intuition, not theorem-proving.

2.5 How to actually build the intuition

Two resources do the heavy lifting, and both are free:

- **3Blue1Brown's *Essence of Linear Algebra*** — genuinely the best visual intuition available for this material. Watch the whole series once; it reframes matrices as transformations far more effectively than any textbook chapter.
- **MIT 18.06 (Gilbert Strang)** — the standard reference course if you want to go deeper on mechanics and proofs.

I also find it helps to *hand-verify* a couple of matrix products on paper with tiny numbers (2×2), just so the shape arithmetic and the "compose two transformations" idea feel owned rather than handed to me.

2.6 The honest scope

You do **not** need to prove theorems about vector spaces. You need three things:

1. Read a matrix multiply in code and know the output shape before running it.
2. Have working intuition for what the operation is doing geometrically.
3. Recognize when something described as "projecting into an embedding space" is just a matrix multiply — because that's what it almost always is.

That's the whole bar, and it's very reachable.

2.7 What to actually learn

- Vectors, matrices, and matrix multiplication as *composed linear transformations*.
- Dot products and what they measure (alignment) — the engine of self-attention.
- Tracking tensor shapes through a sequence of operations **without running the code**.
- Eigenvalues/eigenvectors at an intuitive level ("natural directions").

2.8 References

- 3Blue1Brown, *Essence of Linear Algebra* — the best visual intuition available.
- MIT OpenCourseWare 18.06, *Linear Algebra* (Gilbert Strang).

2.9 Reflection

"Project the patch embeddings into the language model's embedding space" — draw, in your own words, what that is doing mechanically, using only "matrix," "transformation," and "vector."

2.10 Try this week

Take a 2×2 matrix, multiply it by a few different vectors on paper, and plot (or mentally visualize) how each vector moves. Find one matrix that rotates, one that stretches, and one that projects (collapses a dimension). Now — bonus — write the same thing in NumPy: a matrix, a batch of vectors, one `@`, and a `print` of the shapes. That's the whole week.

Chapter 3

CH03 • Calculus & Optimization

How a model actually learns: gradients, and walking downhill.

3.1 Getting here

Path: start here — Programming for ML and Linear Algebra make it easier.

3.2 Premise

Training any model in this document — from a two-line logistic regression to a trillion-parameter transformer — is the same basic act: define a **loss** that measures how wrong the model is, compute the **gradient** of that loss with respect to every parameter (which direction would make it worse), and nudge every parameter a small step in the opposite direction.

3.3 Gradient descent is the whole story

Let's make that concrete, because "gradient descent" is thrown around enough that it can become a box-label rather than a mechanism. A parameter has a value; changing it changes the loss. The gradient tells you, for each parameter, *the direction in which the loss increases fastest*. You want to decrease the loss, so you move each parameter a little bit in the *opposite* direction. Repeat. That's it.

That's training. The "learning rate" is just how big each step is. Everything else — Adam, momentum, learning-rate schedules — is a refinement of the step size and direction, not a new idea.

3.4 The calculus is narrower than a full course

You do not need a full calculus sequence. You need three ideas:

- **Derivatives** — the rate of change of a function: the slope at a point.
- **Partial derivatives** — the same idea, but for a function of many variables: change *one* variable, hold the rest fixed.
- **The chain rule** — how to compute the derivative of a *composed* function by multiplying the derivatives of its pieces.

That last one is the sleeper. **Backpropagation** — the algorithm that actually computes gradients in a neural network — is nothing more than a systematic, efficient application of the chain rule through a deeply nested composition of simple functions, working backward from the loss to every parameter. When someone says "backprop," they mean: walk the composition backward, applying the chain rule at each step. Once you've written a forward pass, backprop is "run it in reverse, multiplying slopes."

3.5 Why high-dimensional optimization is genuinely hard

Beyond the mechanics, it's worth building real intuition for *why* training isn't just "run gradient descent and win." Loss landscapes for large models are **not simple bowls**. They have:

- **Flat regions** where gradients vanish and progress stalls — you're on a plateau, and a plain fixed step size may never get you off it.
- **Saddle points** — directions where some slopes go up and others down, so gradient descent can crawl to a stop even though it's not at a minimum.
- **Sharp, narrow valleys** — where a too-big step overshoots and bounces.

This is *why* practical optimizers do more than plain gradient descent. **Momentum** remembers past directions and rolls over small flat spots. **Adam** scales each parameter's step size based on recent gradient magnitudes, which handles the wildly-different scales that appear in big models. And it's why **learning rate** — how big a step you take — is one of the most consequential hyperparameters in any training run you'll encounter later in this map. Too big: you overshoot and diverge. Too small: you crawl and never finish. Getting a feel for that knob is a genuinely valuable skill.

3.6 How to make it stick

3Blue1Brown has two series that together make this click: *Essence of Calculus*, and a dedicated *Backpropagation* video that shows the chain rule working through a real network. Stanford CS231n's "*Backpropagation, Intuitions*" notes are the standard written companion. But the thing that really lands it is the same advice as Chapter 01: **hand- roll a gradient descent** for something small (linear regression, or a tiny 2-layer net) and watch the loss fall. The moment you've done that, "the model learns by gradient descent" stops being a slogan.

3.7 What to actually learn

- Derivatives and partial derivatives as rates of change.
- The chain rule, and why it's the mechanical core of backpropagation.
- Gradient descent: compute a gradient, then step opposite to it.
- Why learning rate matters, and a working intuition for non-convex loss landscapes (plateaus, saddle points, narrow valleys).

3.8 References

- 3Blue1Brown, *Essence of Calculus*.
- 3Blue1Brown, *Backpropagation*.
- Stanford CS231n notes, *Backpropagation, Intuitions*.

3.9 Reflection

Explain backpropagation to someone who knows what a function is but has never heard of neural networks, using only "nested functions," "slope," and "going backward." If you can't, re-watch the 3Blue1Brown backprop video.

3.10 Try this week

Take the linear-regression-by-hand from Chapter 01's exercise and add **momentum** to the update, then **Adam**. Watch how the loss curve changes on a deliberately ill-conditioned problem (say, features on wildly different scales). Bonus: plot the loss for three learning rates — too small, good, and too big — and *see* the divergence.

Chapter 4

CH04 · Probability & Statistics

The language every model's uncertainty, and every training objective, is written in.

4.1 Getting here

Path: start here.

4.2 Premise

Almost every model in this document doesn't output a single deterministic answer — it outputs a **probability distribution**, and the training objective is almost always some form of *"make the true answer more likely under the model's distribution."*

4.3 The vocabulary to build first

Before the deep ideas, you need working definitions for the basics, because everything builds on them:

- A **random variable** — a value drawn from some distribution (e.g., "the next token").
- A **probability distribution** over its possible values.
- A **joint distribution** — the probability of several things happening together.
- A **conditional distribution** — the probability of one thing *given* that another is already known.
- **Bayes' theorem** — the rule relating the two directions of conditioning to each other ($P(A|B)$ to $P(B|A)$). You'll meet this again in Part III when "we observe evidence and update a belief" comes up.

4.4 Two ideas that do outsized work throughout this map

Two specific concepts recur so much that they're worth real fluency, not just recognition:

Cross-entropy and KL divergence. Cross-entropy measures how well a predicted probability distribution matches the true one — it's the *actual loss function behind next-token prediction* in every language model in Part I. When a model "predicts the next token," the training signal is cross-entropy between its predicted distribution and the correct token. KL divergence measures how *different* two distributions are, and it shows up explicitly in RLHF (Part I, Stage 6 / Part III, Stage 5) as the penalty term that keeps a fine-tuned model from drifting too far from where it started. The intuition: cross-entropy is "how surprised am I by reality, given my model?" and KL divergence is "how much extra surprise does my model add over the truth?"

Expectation and variance. An **expectation** is a probability-weighted average. This one sentence is worth more than it sounds: essentially every "policy gradient" or "expected reward" expression in Part III is, underneath the notation, exactly that — an average over possible outcomes, weighted by how likely each is under the current policy. Once you see "expected reward" as "weighted average over what might happen," Part III loses most of its mystery.

4.5 The statistics habits that actually matter

On the statistics side, the practically important things are *habits*, not theorems:

- **Understand overfitting in terms of noise.** A model can fit the *sample* noise rather than the true underlying pattern. This will be the single most recurring failure mode in the entire book.
- **Trust only held-out data.** A held-out test set is the only honest way to estimate how a model performs on new data. Everything you see on the training set is suspect.
- **Be skeptical of small samples.** If a result is reported from a tiny number of trials, the variance is enormous and the result means little. This skepticism is a career-long skill in ML, and it matters more than almost any formula.

4.6 The practical scope

You need enough probability to *think* in distributions and enough statistics to be an honest evaluator. Two free resources get you there:

- **Seeing Theory** (Brown University) — a visual, interactive introduction to probability and statistics. Ideal for building intuition without a wall of notation.
- **Khan Academy's Statistics & Probability** — solid, structured, and free.

4.7 What to actually learn

- Random variables, probability distributions, and conditional probability.
- Cross-entropy and KL divergence — and *where each shows up later in this map* (cross-entropy in next-token prediction; KL in RLHF).
- Expectation as a probability-weighted average — especially for policy gradients (Part III).
- Overfitting, train/test splits, and basic sample-size skepticism.

4.8 References

- Seeing Theory (Brown University).
- Khan Academy, Statistics & Probability.

4.9 Reflection

A model "predicts the next token." In your own words, what is the *training signal* measuring, and why is it a statement about probabilities rather than a single right answer?

4.10 Try this week

Take a tiny dataset, fit a high-capacity model (a deep net or even a high-degree polynomial) and a low-capacity one. Plot training error vs. test error for both as you train longer. *See* overfitting happen: the high-capacity model's training error keeps falling while its test error rises. That's the whole lesson of this chapter, in one plot.

Chapter 5

CH05 • Core Machine Learning Concepts

What "training a model" means, before any of it involves a transformer.

5.1 Getting here

Path: Programming for ML → Linear Algebra → Calculus → Probability → **here**.

5.2 Premise

This prerequisite stage is about the conceptual vocabulary of machine learning itself — the ideas Parts I through III will use constantly without re-explaining. It's the vocabulary; the network those ideas train is the next chapter.

5.3 The anatomy of training

A **dataset** is typically split three ways, and each part serves a distinct job:

- **Training data** — what the model learns from.
- **Validation data** — used to tune choices like learning rate *without* contaminating the final evaluation. You try several settings, pick the one that does best on validation, and only then face the test set.
- **Test data** — touched *only at the very end*, to honestly estimate real-world performance.

The reason for the split is that any tuning done on a dataset leaks information about it. A model evaluated on data it was tuned against looks better than it actually is. The test set is the one honest exam.

A **loss function** quantifies how wrong a model's prediction is for a given example. "Training" is mechanically using the calculus and optimization from Stage 3 to adjust parameters to make the *average loss on the training set* small.

5.4 Overfitting: the one failure mode to know deeply

If there's a single concept to understand deeply in this whole part, it's **overfitting**: a model with enough capacity can simply *memorize* its training data — achieving near-perfect training loss while performing badly on anything new, because it learned the *specific noise* in that dataset rather than the underlying pattern that generalizes.

This is not an edge case. It is the fundamental tradeoff of the entire field, and every model discussed later in this book — a scaled-up transformer, an RL policy, a video generator — is fighting the same underfitting/overfitting tradeoff, just at far larger scale and with far more sophisticated tools.

Regularization techniques exist specifically to fight it:

- **Weight decay** — penalizes large weights, encouraging the model to use simpler solutions.
- **Dropout** — randomly turns off neurons during training, forcing the network not to rely on any one pathway.
- **Early stopping** — stop training when validation error starts rising even as training error falls.

If you only ever remember the shape of the problem — "capacity vs. generalization, held out until the end" — you've got the spine of this chapter.

5.5 From vocabulary to an actual network

That's the spine. The object this vocabulary trains — a stack of layers, run forward and then back-propagated — now has its own chapter: **Stage 6, Neural Networks**. That's where you build it by hand. Everything later in the map is a variation on that object, so it's worth owning before you meet a transformer.

5.6 The meta-lesson

Notice how much of this part was *not* about any specific architecture. That's deliberate. Parts I, II, and III are all just different ways of applying the same spine: define a loss, compute gradients, nudge parameters, watch out for overfitting. If the spine is solid, every later chapter is an elaboration on it rather than a leap.

5.7 What to actually learn

- Train / validation / test splits, and why each serves a different purpose.
- Loss functions as a quantified measure of "how wrong" a prediction is.
- Overfitting vs. underfitting, and regularization (weight decay, dropout, early stopping) as the standard countermeasure.
- The conceptual spine of training — the network itself (layers, forward pass, backprop) is Stage 6.

5.8 References

- Goodfellow, Bengio & Courville, *Deep Learning* — free online, the standard reference text.
- fast.ai, *Practical Deep Learning for Coders* — a hands-on, code-first companion to the theory.
- Stanford CS231n course notes — excellent on fundamentals.

5.9 Reflection

You have a model that scores 99% on training data and 70% on test data. Diagnose it in one sentence, and name two levers you'd pull.

5.10 Try this week

Build a tiny feedforward network from scratch in NumPy (forward pass + backprop by hand), train it on a toy classification problem, and watch it overfit: crank the capacity up and down and compare train vs. test accuracy. (The full build-a-network walkthrough is Stage 6.)

Chapter 6

CH06 • Neural Networks (MLPs, Forward & Backward)

Putting the pieces together: layers, the forward pass, backprop, and a training loop.

6.1 Getting here

Path: Programming for ML → Linear Algebra → Calculus → Probability → Core ML → **here**. If the math already feels solid, this is where the real learning starts — and everything in Parts I–III is a variation on what you build here.

6.2 Premise

A neural network is the object every later stage in this map is a variation on: a stack of layers, trained end-to-end by backpropagation. Get this once, by hand, and attention, RL policies, and world models stop looking like new species and start looking like elaborations.

6.3 Why we need nonlinearity

You already know a linear model: linear regression is one weight vector and a bias, and its prediction is just $W \cdot x + b$. Here's the catch that motivates everything else — **stack two linear transformations and you still have a linear transformation** (a matrix times a matrix is another matrix). No matter how many linear layers you chain, the whole thing collapses back into a single linear map. A deep stack of linear layers would be no more expressive than one.

So a network needs something *nonlinear* between the layers. That's the entire reason "deep" learning is possible.

6.4 A layer: transform, then nonlinearity

A single layer is two steps:

1. **A linear transformation** — a matrix multiply plus a bias ($W \cdot x + b$), exactly the matrix multiplication from Stage 2.
2. **A nonlinearity** — a simple function applied elementwise, like **ReLU** ($\max(0, x)$), sigmoid, or GELU.

That elementwise nonlinearity is the crucial part: it breaks the collapse, letting each layer add something a linear map couldn't. Stack these — transform, nonlinearity, transform, nonlinearity, ...

— and you get a **multilayer perceptron (MLP)**, the plain feedforward network that everything else builds on. "Deep" just means "many layers."

6.5 The forward pass

Run an input through the stack and you get an output. Concretely:

```
x  -> layer1 (W1·x + b1, ReLU)  -> h1
    -> layer2 (W2·h1 + b2, ReLU) -> h2
    -> output layer (W3·h2 + b3) -> prediction
```

Mechanically, the forward pass is nothing more than a sequence of matrix multiplications and simple elementwise functions — which is why Stage 2's linear algebra is the notation the whole field is written in. If you can track the shapes through that chain, you understand the forward pass.

6.6 The loss

To train, the network needs a single number saying how wrong it currently is: the **loss** (from Stage 4). For classification it's usually **cross-entropy** between the predicted probabilities and the true label; for regression, mean squared error. The loss turns "how wrong" into a differentiable quantity — which is what lets us compute a gradient.

6.7 Backpropagation

This is the heart of the chapter, and the single home for it. **Backpropagation is the chain rule (Stage 3) applied systematically through the network, working backward from the loss to every parameter.**

Here's the whole idea. The forward pass is a deeply nested composition of functions: `loss = f3(f2(f1(x)))`. The chain rule says the derivative of a composition is the product of the derivatives of its pieces. So to find how the loss changes when you nudge a weight deep inside the network, you multiply the local derivatives along the path from that weight out to the loss. Backprop just does that efficiently:

1. **Forward pass** — compute each layer's output, keeping the intermediate values.
2. **Backward pass** — starting at the loss, compute the gradient with respect to the last layer, then pass it backward to the previous layer, multiplying by each layer's local derivative as you go — all the way to the first weight.

The magic is reuse: each layer's gradient computation feeds the previous layer's, so you get the gradients for *every* parameter in one backward sweep rather than one pass per parameter. This is why training a network with millions of weights is feasible at all.

If you want to *feel* it rather than nod at it, the classic move is to implement it by hand for a two-layer net (see *Try this week*) — it's tedious exactly once, and afterward "the network learns by gradient descent" is something you've literally done, not a slogan.

6.8 The training loop

Backprop gives gradients; an **optimizer** uses them to nudge the weights. The loop:

1. **Forward** — compute predictions for a **batch** of examples.
2. **Loss** — measure how wrong they are.

3. **Backward** — compute gradients of the loss w.r.t. every parameter.
4. **Step** — update each parameter a little in the opposite direction of its gradient.

Then repeat over the dataset (an **epoch** is one full pass). The three knobs that matter most:

- **Learning rate** — the step size. Too big and training diverges; too small and it crawls. (This is the concrete payoff of Stage 3.)
- **Optimizer** — plain SGD is the baseline; **momentum** helps roll through flat spots; **Adam** adapts a step size per parameter and is the usual default.
- **Batch size** — how many examples per update; a tradeoff between noisy-but-fast and stable-but-slow.

None of this is exotic: it's the gradient-descent loop from Stage 3, now with many parameters and a smarter step rule.

6.9 Capacity, and why we hold data out

The recurring failure mode is **overfitting** (covered in Stage 5, and worth re-reading here because you'll now see it in action): a network with enough capacity can memorize its training data and still fail on anything new. The defenses are the same — **weight decay**, **dropout**, **early stopping** — and the honest measure is always a **held-out test set** you touch only at the end. In practice you tune on a **validation** split and report on test. Watching your own tiny net overfit, then stop overfitting when you add these, is the lesson.

6.10 The meta-point

Everything later in this map is this chapter, scaled up and specialized: a transformer is a particular arrangement of layers plus a particular loss, an RL policy is a network with a different objective, a world model is a network trained to predict. If the spine here is solid — forward, loss, backward, step — then no later architecture is a leap, only an elaboration.

6.11 What to actually learn

- Why stacking linear layers collapses, and why a nonlinearity between them is what makes depth expressive.
- A layer as *linear transformation + elementwise nonlinearity*, and an MLP as a stack.
- The forward pass as a chain of matrix multiplies plus nonlinearities.
- **Backpropagation** as the chain rule run backward through that chain — and why reuse makes it efficient.
- The training loop (forward → loss → backward → step), and the roles of learning rate, optimizer, and batch size.
- Capacity vs. generalization: overfitting, regularization, and why the test set is held out.

6.12 References

- Goodfellow, Bengio & Courville, *Deep Learning* — free online; chapters 6–8 cover this exactly.
- 3Blue1Brown, *Backpropagation* — the clearest visual derivation of the backward pass.
- Karpathy, *micrograd* and *Neural Networks: Zero to Hero* — build autograd and an MLP from scratch.
- fast.ai, *Practical Deep Learning for Coders* — the hands-on companion.

6.13 Reflection

Why does a stack of linear layers collapse into a single linear map, and what exactly does adding a ReLU between them change? Then: why can backprop compute the gradient for *every* parameter in one backward pass instead of one pass per parameter?

6.14 Try this week

Build a small MLP from scratch in NumPy: a couple of linear layers, a ReLU, a softmax cross-entropy loss, and **your own backprop** (derive the gradients by hand, don't let a framework do it). Train it on a toy two-class problem and watch the loss fall. Then crank the capacity up and down and compare train vs. test accuracy — that's overfitting you can see, on a network you built and backpropped yourself. This is the single most valuable "I actually understand this" exercise in all of Part 0.

This closes Part 0. The next line begins where this leaves off — by asking how to make sequence models stop reading one word at a time.

Part II

Part I — Modern LLM Architectures

Contents of this part

- CH01 · Recurrent Sequence Models (RNN & LSTM)
- CH02 · Attention & Self-Attention
- CH03 · The Transformer Block
- CH04 · Pretraining Objectives & Tokenization
- CH05 · Scaling, Efficiency & Long Context
- CH06 · Alignment: RLHF & DPO
- CH07 · Multimodal LLMs
- CH08 · Agents & Tool Use

From recurrence to agents that plan and call tools — eight stages, in the order that makes each one make sense.

This is the **language line** of the map. It's the longest-running thread of the past decade in AI, and it's where the modern era actually began. Part 0 left you able to build and train a feedforward network; this part asks the question that turns that into the world of large language models: *how do you make a model handle sequences — text, in particular — without reading them one word at a time, and hopelessly compressing everything it's seen into a single fixed state?*

The arc of the part:

1. **Recurrent Sequence Models (RNN & LSTM)** — the *interchange foundation*: the sequence machinery language built on before attention, and still the live temporal model in applied video/action recognition.
2. **Attention & Self-Attention** — the mechanism that broke recurrence (the *fix* for Stage 1).
3. **The Transformer Block** — the architecture built around attention, and an *interchange with computer vision*.
4. **Pretraining Objectives & Tokenization** — what turns raw text into a training signal.
5. **Scaling, Efficiency & Long Context** — what changes when everything gets bigger.
6. **Alignment: RLHF & DPO** — an *interchange with deep RL*; how you make a raw next-token predictor follow instructions.
7. **Multimodal LLMs** — an *interchange with video CV*; feeding images and video in.
8. **Agents & Tool Use** — from answering in one shot to planning and acting over many steps.

The through-line is a single idea you'll meet at Stage 2 and never fully leave: a transformer treats a sequence as a set of tokens that mutually attend to one another. Everything after that — scaling it, aligning it, making it multimodal, making it act — is a consequence of that one design decision.

And here's the meta-point I want you to keep in view the whole time: **this line ends exactly where the other two begin**. Stage 8's "agents that loop over actions" is the bridge to Part III's action line, and Stage 7's "vision tokens in a language model's context" is the bridge to Part II's vision line. Keep the map in mind and no chapter will feel like it's off on its own.

Let's start where sequence modeling started — with the recurrent machinery that handled sequences before attention came along.

Chapter 7

CH01 • Recurrent Sequence Models (RNN & LSTM)

The machinery that modeled sequences before attention — and that applied temporal modeling still runs on.

7.1 Getting here

Path: Part 0 (especially Core ML and Neural Networks) → **here**. This opens the language line.

7.2 Interchange — with Video CV, Stage 7 (Action & Activity Recognition)

X1 (LLM 1 ↔ Video CV 7). Sequence data — language, then time — is the same problem in two fields, and recurrence is the machinery both reached for. From the language side: recurrence is how we modeled sequences before attention fixed its flaws (the next chapter). The video line still runs on it where a compact temporal signal needs memory — the LSTM inside Action & Activity Recognition. Read that chapter (**Part II, Stage 7, CH07**) for the applied, video-facing half of the same idea.

7.3 Premise

You know how to build a neural network that maps one fixed input to an output — a feedforward net (an MLP). But language and video are **sequences**: variable in length, and ordered in a way that matters. This chapter is about the models built to handle that — **recurrent neural networks (RNNs)** and their far more practical descendants, **LSTMs** and **GRUs**. They were the standard sequence model until attention came along, and they're still the honest answer in a lot of applied temporal work.

7.4 Why feedforward nets can't do sequences

The MLP you built in Part 0 has two hard limits for sequence data:

1. **Fixed input.** It expects one input of a fixed size. A sentence can be three words or thirty; an action can be a half-second or five — a network with a fixed input width can't take them.
2. **No order, no memory.** Even if you padded everything to the same length, an MLP treats each input independently and has no notion of "this came before that." But in language, "dog bites man" and "man bites dog" are different — order is the whole point. And much of meaning lives in *history*: "he" refers to something earlier.

You could, in principle, feed the whole sequence in at once and let the network figure out order from position — that's essentially what the *attention* chapter (next) will do — but that gets expensive and ignores a simpler idea: **process the sequence one step at a time and carry what matters forward in a hidden state.**

7.5 The RNN: a network with a hidden state

The core idea of a recurrent network is to **reuse the same network at every timestep, feeding it both the current input and its own previous output.** At each step, the network takes the current token and the previous hidden state, and produces a new hidden state (plus, if we're generating, a prediction). In symbols, one learned recurrence rule maps the current input and old hidden state to the new hidden state:

$$h_t = \text{activation}(W_x x_t + W_h h_{t-1} + b)$$

Because h_t depends on h_{t-1} , which depended on h_{t-2} , and so on, the hidden state at step 50 *contains* (in compressed form) information from all the earlier steps. That's the memory. And because the same weights are used at every step, the network has learned one recurrence rule that applies however long the sequence is — it generalizes to any length, which solves the fixed-input problem.

This is the architecture behind the classic **RNN language model**: at each step, predict the next token from the current hidden state, and step forward. Trained with the same backprop you already know — with one addition.

7.6 Backpropagation through time

Training a recurrent net is just backprop, but you have to unroll the recurrence: because the loss at step 50 depends on h_1 through the chain of hidden states, you backpropagate *through all the timesteps* — **backpropagation through time (BPTT)**. You "unroll" the network into a very deep feedforward net (one copy per timestep, sharing weights) and backprop through the whole thing. If you've done backprop for a deep MLP, BPTT is the same math over the unrolled graph.

7.7 The fundamental problem: vanishing gradients

And here is where recurrence hits its wall, and why "simple" RNNs rarely work in practice. To learn a long-range dependency — say, that a noun near the start of a sentence determines a verb much later — the gradient has to flow back through many timesteps. But each step multiplies the gradient by the recurrent weight matrix. If those weights are even slightly less than 1, the gradient **vanishes** (shrinks to nothing over ~10–20 steps); if slightly more than 1, it **explodes**. Either way, a plain RNN simply cannot learn to remember things across long gaps — its hidden state gets overwritten and its training signal vanishes.

This is exactly the "compression bottleneck" the attention chapter describes: a fixed-size hidden state is a bottleneck, and gradient flow makes it worse. Fixing *this* problem is what the LSTM is for.

7.8 LSTM: gates that decide what to remember and forget

An **LSTM** (Long Short-Term Memory) keeps the same "carry a state forward" structure but adds a second state — a **cell state** (call it c_t) — that runs alongside the hidden state and is specifically built to carry information across long gaps without being overwritten. Three learned **gates** control it:

- A **forget gate** decides how much of the old cell state to keep.

- An **input gate** decides how much new information to write in.
- An **output gate** decides what to expose from the cell state as the new hidden state.

Each gate is a small network with a sigmoid (0–1) that *multiplies* the state, letting the LSTM learn "keep this memory, drop that one, write this in." Because the cell state has a near-identity path (the forget gate can be ~ 1 , so the gradient doesn't vanish as it flows back through the cell), LSTMs can actually learn dependencies over hundreds of steps — the vanishing-gradient problem that crippled plain RNNs is largely solved by *gating*.

A **GRU** (Gated Recurrent Unit) is the same idea with fewer gates (a combined update gate and no separate cell state) — simpler and often just as good.

If you want the cleanest mental picture: **an LSTM is an MLP with a memory cell and three learned on/off dials deciding what to remember, write, and read out at each step.** It's a strictly more capable version of the networks you already know how to build and backprop.

7.9 Why attention replaced it (and why it still matters)

The next chapter tells the full story, but the summary here is: recurrence has two structural costs that attention removes.

1. **A fixed-size bottleneck.** Everything the model knows about the past is squeezed into one hidden vector. Attention instead lets each position *look directly* at every other position, so no information has to survive a bottleneck.
2. **No parallelism.** Because each step depends on the previous one, an RNN must process a sequence left-to-right — you can't parallelize across time. Attention computes all positions at once (it's just matrix multiplies), which is what made training at huge scale possible.

So for *language*, recurrence was superseded by attention starting in 2017. **But** — and this is the honest part that matters for real work — recurrence did not disappear:

- **Applied temporal modeling still runs on it.** When your input is a *short, compact* sequence of features (like pose-keypoint frames in action recognition), an LSTM is cheap, needs little data, and has no bottleneck problem in practice — which is exactly why ActionAI-style systems (Part II, Stage 7) use it. Recurrence excels precisely where attention is overkill.
- **RL uses it for memory.** In partially observable environments (a robot that can't see the full state), an LSTM layer gives the policy a memory of its own history — the standard trick for handling POMDPs.
- **Sequence modeling historically.** RNN language models were *the* approach to language modeling before attention, and every modern framing of "why attention?" assumes you understand what it replaced.

This is what makes recurrence an *interchange* rather than a footnote: it's not just the history of the language line — it's the live machinery of the video line's temporal models and of memory-hungry RL. Understand it once and three tracks of the map make more sense.

7.10 What to actually learn

- Why feedforward nets can't handle sequences: fixed input, no order, no memory.
- The RNN: same weights at every step, a hidden state that carries history forward.
- Backpropagation through time, and the vanishing/exploding gradient problem.

- How LSTM gates (forget/input/output) fix vanishing gradients, and how a GRU simplifies it.
- Why attention replaced recurrence for language (bottleneck + no parallelism), and where recurrence still lives: action recognition (Part II, Stage 7), RL partial observability, and short-sequence temporal work.

7.11 References

- Hochreiter & Schmidhuber, *Long Short-Term Memory* (1997) — the original LSTM.
- Karpathy, *The Unreasonable Effectiveness of Recurrent Neural Networks* — the best hands-on intuition for RNNs and char-level LSTMs.
- Olah, *Understanding LSTM Networks* — the clearest visual walkthrough of the gates.
- Colin Raffel, *A Simple Way to Initialize Recurrent Networks of ReLU and Linear Units* — for the vanishing-gradient fixes that make modern RNN variants work.
- Sutton & Barto (2nd ed.), for where recurrent policies appear in partially observable RL.

7.12 Reflection

Attention is "better" than recurrence for language — but ActionAI-style systems still use LSTMs for temporal modeling. What is it about a *short, compact* sequence that makes the recurrent bottleneck a non-issue there, and why does that break down for long language?

7.13 Try this week

In your study→implement→write style: hand-roll a small RNN in PyTorch and train it to predict the next character in a short text. First watch a *plain* RNN fail to carry information across a long gap, then swap in an LSTM cell and watch the same task succeed. Then *write* a paragraph connecting what you just saw to why attention (the next chapter) was needed for language — and why the LSTM you built is the same cell ActionAI uses on pose-keypoint sequences. That's the whole interchange, demonstrated.

Chapter 8

CH02 • Attention & Self-Attention

The mechanism that let sequence models stop reading one word at a time.

8.1 Getting here

Path: Neural Networks → Recurrent Sequence Models → **here**.

8.2 Premise

Before attention, sequence models were built on **recurrence** (the RNNs and LSTMs of Stage 1) — and recurrence had two fatal problems: a compression bottleneck, and no parallelism.

8.3 Why recurrence was the bottleneck

An RNN or LSTM read a sentence one token at a time, compressing *everything it had seen* into a fixed-size hidden state and passing that state forward. This worked, but it created a hard bottleneck: by the time the model reached word fifty, information from word two had to survive forty-eight steps of compression to still matter. Long-range dependencies — the thing that actually matters in language — got smeared out. It also made training slow, because each step depended on the previous one, so you couldn't parallelize across a sequence. You had to process the sentence sequentially, one token at a time, no matter how many GPUs you had.

Both problems are structural. Recurrence *forces* a single fixed-size summary and *forces* sequential processing. You can't fix them with more compute — you have to change the architecture.

8.4 Self-attention solves both at once

Self-attention's move is conceptually simple and structurally total: **instead of compressing the past into a single state, every position in a sequence directly looks at every other position and decides how much to "attend" to it.**

Mechanically, each token is projected into **three vectors** — a query, a key, and a value:

- The **query** from one token is compared against the **keys** of all other tokens (via a dot product — remember Chapter 02; this is exactly the "how aligned are these two vectors" operation) to produce **attention weights**.
- Those weights are used to compute a **weighted sum of the value vectors**.

The result is a new representation of that token that has *directly pulled in relevant context from anywhere in the sequence, in a single step, with no compression bottleneck*. Word two's information can reach word fifty without surviving forty-eight steps of sequential smearing — because word fifty can attend straight back to word two.

8.5 The parallelism that made the era possible

Here's the detail that turned this from a nice idea into a revolution: **every token's attention computation is independent of the others**. Because each token computes its own query, key, and value from the input, the whole operation is a handful of matrix multiplications — and matrix multiplications parallelize beautifully on GPUs.

This is why the transformer era happened. Attention isn't just more expressive than recurrence; it's dramatically more *trainable at scale*. Recurrence limited you to sequential processing; attention turned sequence modeling into something you could run across thousands of GPUs in parallel.

8.6 Multi-head attention

A single attention pass captures one kind of relationship. **Multi-head attention** runs several attention operations in parallel with *different learned projections*, so the model can track several different kinds of relationship at once — syntax, coreference, topical relevance. Each "head" gets to specialize. This is why transformers can hold a rich picture of a sentence: not one attention pattern, but many, attending to different things.

8.7 The masks that change behavior

There's a crucial dial on attention: **masks**. Because every token can see every other, you have to *choose* what a token is allowed to see:

- **Causal (autoregressive) masking** — each token may only attend to tokens before it. This is what makes a model generative: it can only predict the next token from the past. This is what nearly every modern LLM uses.
- **Bidirectional masking** — each token attends to both left and right context. Better for understanding tasks (like BERT-style classification), but not naturally generative.

The distinction matters enormously and reappears constantly, especially when you meet encoder-vs. decoder-only designs in the Transformer Block chapter (Part I, Stage 3) and in every generative model in this book. It's also the same mechanism computer vision borrows for its Vision Transformers — same operation, different tokenization, text tokens in one field and image patches in the other — which is the Transformer ↔ ViT interchange you'll meet at Part II, Stage 2 and revisit all through Part II.

8.8 What to actually learn

- The query/key/value formulation and **scaled dot-product attention** (the dot product measures alignment; the scaling keeps the softmax from saturating as vectors grow).
- Why attention parallelizes across a sequence while RNNs cannot.
- Multi-head attention and what different heads tend to specialize in.
- Attention masks: causal (autoregressive) vs. bidirectional.

8.9 References

- Vaswani et al., *Attention Is All You Need* (2017) — the paper that introduced the transformer.
- Jay Alammar, *The Illustrated Transformer* — the best visual walkthrough of the mechanism.
- *The Annotated Transformer* — the paper reimplemented line by line in PyTorch.

8.10 Reflection

Why did recurrence have *two* structural problems, and how does self-attention resolve *both*? (Hint: one is about information, the other about training speed.)

8.11 Try this week

Implement scaled dot-product attention from scratch in PyTorch — a matrix of queries, keys, values, one dot product, a softmax, a weighted sum — and verify the output shapes by hand. Then build it twice: once with causal masking, once bidirectional, and feed both a short sequence. That’s the entire mechanism, and now you own it.

Chapter 9

CH03 • The Transformer Block

The full architecture — and the point where language and vision start using the same machinery.

9.1 Getting here

Path: Recurrent Sequence Models → Attention → **here**.

9.2 Interchange — with Video CV, Stage 2 (Vision Transformers)

X2 (LLM 3 ↔ Video CV 2). This block — multi-head self-attention plus a position-wise feed-forward layer, wrapped in residuals and layer norm — is the same architecture the vision line points at image patches to build its ViT. From the language side it's the backbone of every LLM; the vision half is identical machinery fed a different tokenizer. Read **Part II, Stage 2, CH02 (Vision Transformers)** alongside.

9.3 Premise

Self-attention is one operation; the **transformer** is the architecture built around it. And it's the single most important thing in this entire book, because variants of it appear at every later stage of the map — including, in almost identical form, in the vision and reinforcement-learning lines.

9.4 What the block actually is

A transformer block interleaves two things, each wrapped in a residual connection and layer normalization:

1. **Multi-head self-attention** — from Chapter 21. This is where tokens exchange information with each other.
2. **A position-wise feed-forward network** — a small MLP applied independently to each token. This is where each token does its own nonlinear thinking.

Stacking a dozen to a few hundred of these blocks, with an **embedding layer** at the bottom (turning token IDs into dense vectors) and a **projection to vocabulary logits** at the top, is essentially the entire architecture behind every major language model.

The residual connections (a skip from input to output, added back) and layer normalization are the unsung heroes: they're what let you stack *hundreds* of blocks without the training collapsing. They keep gradients flowing and activations stable. This is the same residual trick you met in Part 0's feedforward nets, applied so the depth itself becomes possible.

9.5 Two design choices that matter enormously

1. Positional information. Attention has no inherent sense of order — it treats a sequence as a *set* of tokens that mutually attend to one another. Swap two tokens and attention treats them identically. So position has to be injected explicitly. The options evolved:

- Original **sinusoidal encodings** (from the 2017 paper) — fixed, hand-designed.
- **Learned position embeddings** — trained like any other parameter.
- **Rotary position embeddings (RoPE)** — the current default in most serious models. It *rotates* query/key vectors as a function of position, and crucially generalizes better to sequence lengths not seen during training (more on that in Chapter 24).

2. Encoder vs. decoder vs. encoder-decoder.

- **Encoder-only** — sees the whole sequence at once (bidirectional). Great for understanding tasks: classification, embeddings. BERT is the archetype.
- **Decoder-only** — causal masking, generates left to right. This is what nearly every modern LLM uses. The generative default.
- **Encoder-decoder** — translation-style, input sequence mapped to output sequence.

The modern story is that decoder-only won for generation, and most frontier LLMs are decoder-only transformers trained to predict the next token.

9.6 The interchange: vision reads the same block

Here's the payoff of Chapter 21, and the heart of the whole map's thesis. The **Vision Transformer (ViT)** takes an image, cuts it into fixed-size patches (say 16×16 pixels), flattens and linearly projects each patch into a **token embedding**, adds a position embedding, and then runs the *exact same transformer block stack used for text*. There is no vision-specific convolutional inductive bias left in the core computation — a ViT genuinely is a language-model architecture pointed at a different tokenization scheme.

This is not an analogy. It's the same machinery. Which is exactly why progress in one field transfers so directly to the other: FlashAttention, RoPE variants, and training-stability tricks discovered for LLMs apply to ViTs almost unchanged, and vice versa. When you read that a "multimodal" model (Chapter 26) feeds image patches and text tokens into the same attention, this is the mechanism making it possible.

9.7 Why this is the highest-leverage thing in the book

Understanding this block deeply — ideally by **implementing one from scratch** — is the single highest-leverage thing you can do across all three tracks in this map. Karpathy's **nanoGPT** is the classic way: build a decoder-only transformer from scratch in a few hundred lines. Do that once, and every later chapter — scaling, RLHF, multimodal, world models, VLA agents — is describing a variation on machinery you've already built.

9.8 What to actually learn

- Encoder-only vs. decoder-only vs. encoder-decoder designs, and when each is used.
- Residual connections and pre-norm vs. post-norm layer normalization (pre-norm became the modern default for stability).

- Positional encoding schemes, especially **RoPE**.
- How the ViT patchifies an image into a token sequence (the interchange).

9.9 References

- Karpathy, *nanoGPT* — build a decoder-only transformer from scratch in a few hundred lines.
- Su et al., *RoFormer* — Rotary Position Embedding (2021).
- Dosovitskiy et al., *An Image is Worth 16x16 Words* — the ViT (2020).

9.10 Reflection

An image is "cut into patches and fed to the same transformer as text." What does that claim imply about what images and text *are*, at the level the transformer sees them?

9.11 Try this week

Build a tiny transformer block from scratch in PyTorch — embedding, a couple of self-attention + feed-forward blocks with residuals and layer norm, and a causal mask — and train it to generate a few characters of text (nanoGPT-lite). Then, bonus, feed it the same sequence of "tokens" where the tokens are image patches (flatten 8×8 blocks of a small image). Watch it not care about the difference. That's the interchange, made literal.

Chapter 10

CH04 · Pretraining Objectives & Tokenization

What actually turns raw text into a training signal.

10.1 Getting here

Path: Attention → Transformer Block → **here**.

10.2 Premise

A transformer block is just a function. What makes it a *language model* is the **objective** it's trained on — the rule that turns raw text into a gradient signal.

10.3 The objective that won: causal language modeling

The dominant choice today is **causal language modeling**: given every token before position t , predict token t . That's it.

What makes this so powerful is that it's *self-supervised* — there are **no labels beyond the text itself**. The text is its own supervision: hide the next word, make the model guess it. Yet training this deceptively simple objective at scale on enough diverse text produces models that pick up grammar, facts, reasoning patterns, and style *purely as a side effect* of getting good at "what word comes next." That's the quietly astonishing result that launched the modern era — language models learn the structure of the world because predicting the next word well requires it.

The alternative, **masked language modeling** (BERT's approach), hides random tokens and asks the model to reconstruct them using *both* left and right context. It produces strong bidirectional representations — better for classification and embeddings — but isn't naturally generative, since it never learns to produce text left-to-right. The pragmatic outcome of the past few years: **decoder-only + causal LM won** for the generative tasks that dominate, while masked/encoder models remain useful for embeddings.

10.4 Tokenization: text has to become numbers first

Before any of this can happen, text has to become numbers. And the *way* it becomes numbers matters a lot more than most people expect.

The naive options both fail. A **word-level** vocabulary can't handle novel or rare words (it sees "*nuclearfission*" and has no token). A **character-level** one makes sequences enormously long and throws away the regularity of common subwords. The standard solution is **subword tokenization**, specifically **byte-pair encoding (BPE)**:

1. Start from individual bytes or characters.
2. Iteratively merge the *most frequent adjacent pair* into a single token.
3. Repeat until you've built a vocabulary of common subword chunks.

The result: common words become single tokens; rare words get split into a few meaningful pieces. This is genuinely consequential — **tokenizer choice affects how efficiently a model handles different languages, code, and numbers**, and quirks in tokenization are behind several well-known LLM failure modes. The classic one: a model struggling to count the letters in a word, because it never sees individual letters as separate tokens. Understanding BPE explains a whole category of seemingly-weird model behavior.

10.5 The rest of the plumbing

Around the transformer core sit two thin pieces you've already met in outline:

- The **embedding layer** converts token IDs into dense vectors (the bottom of the stack).
- The final layer projects the transformer's output back into a **probability distribution over the vocabulary** (the top).

Everything in between is what Chapters 21 and 22 covered. Once you've got the objective, tokenization, and those two projections, you've got a complete language model — the rest of Part I is about scaling it, aligning it, making it multimodal, and making it act.

10.6 What to actually learn

- Causal vs. masked language modeling objectives, and why decoder-only won for generation.
- How byte-pair encoding builds a subword vocabulary.
- Why tokenization choices cause specific, predictable model failure modes (e.g., letter-counting).
- Embedding spaces and what "context window" actually means mechanically (the max sequence length the model can attend over).

10.7 References

- Devlin et al., *BERT* (2018).
- Brown et al., *Language Models are Few-Shot Learners* — the GPT-3 paper (2020).
- Karpathy, *Let's build the GPT Tokenizer* — a hands-on BPE walkthrough.

10.8 Reflection

"Language models learn grammar and facts as a side effect of predicting the next word." Why is that less surprising than it sounds, once you think about what predicting the next word well actually requires?

10.9 Try this week

Train a BPE tokenizer (or use one) on a few different corpora — say English prose, Python code, and a mix of languages — and inspect the vocabularies. See how token *frequency and coverage* differ, and find one example of a word that's a single token in one corpus and split in another. Then explain, in a sentence, why tokenizer choice is a real design decision and not a footnote.

Chapter 11

CH05 · Scaling, Efficiency & Long Context

What changes when the model, the data, and the context window all get bigger.

11.1 Getting here

Path: Transformer Block → Pretraining → **here**.

11.2 Premise

One of the most consequential empirical findings in the field is that transformer performance improves *smoothly and predictably* as you scale up model parameters, dataset size, and compute — as long as you scale them together in the right ratio.

11.3 Scaling laws: how compute should be spent

The original **Kaplan et al.** scaling laws and the later **Chinchilla** paper both fit power-law curves to loss as a function of these quantities, but reached different *practical* conclusions. Chinchilla's famous result: many earlier large models were **undertrained relative to their size** — and that for a fixed compute budget, a *smaller* model trained on *more* data beats a larger model trained on less. This single result reshaped how labs allocate compute.

But the roadmap is now stale here, and the correction is important. Post-Chinchilla research and industry practice moved the goalposts in two ways:

- **Inference cost changes the optimum.** "Beyond Chinchilla-Optimal" (2024) showed that once you count the FLOPs used at *inference* (which, at deployment, dominate), the optimal model is *smaller and more overtrained* than Chinchilla's training-only optimum.
- **Overtraining became standard.** Industry runs 3–20× beyond compute-optimal: Llama-3 8B trained on 15T tokens (~17× Chinchilla), Qwen2.5 on 18T. "Language Models Scale Reliably with Over-Training" (ICLR 2025) showed roughly 4× overtraining is safe. Even Chinchilla's own fitting method is under methodological attack (2026).

So: learn Chinchilla's *logic* (compute-optimal ratios exist), but treat its specific numbers as a moving target — the field now deliberately overtrains for deployment efficiency.

11.4 Efficiency engineering: MoE, FlashAttention, and memory

Scaling forces engineering problems that matter as much as the architecture itself. Three ideas recur constantly:

- **Mixture-of-Experts (MoE).** Replace the single feed-forward block in each layer with many parallel "expert" sub-networks and a learned router that sends each token to only a *few* of them. Total parameter count grows far faster than the compute needed for any single forward pass. This is no longer an exotic trick (Switch Transformers) — it's the frontier standard: DeepSeek-V3 (671B total / 37B active), Llama-4, Qwen3, Kimi K2, Grok all ship MoE at massive scale.
- **FlashAttention.** Rewrites the attention computation to minimize slow memory reads/writes on the GPU rather than reducing floating-point operations — large real-world speedups without changing the math. The lesson: at scale, *memory bandwidth often matters more than raw compute*. FlashAttention-3, vLLM's PagedAttention, and KV-cache tricks are baseline infrastructure now, not research curiosities.
- **KV-cache compression.** During inference the model caches past keys/values; this cache is often the bottleneck. Grouped-query attention (sharing K/V across heads) and **DeepSeek's Multi-head Latent Attention (MLA)** — a low-rank joint compression that cuts the KV cache $\sim 10\times$ — are how long-context models actually stay runnable. (Related: quantizing weights trades a little accuracy for large memory/inference savings.)

A 2023 framing would also mention the Mamba/state-space-model "transformer killer" wave. The 2024–2025 verdict: **hybrids won**. Pure SSMs still lag transformers on in-context recall at scale; the pattern that shipped is interleaving SSM/linear layers with attention (and MLA/sparse attention) where long context dominates. "Attention vs. SSM" turned out to be a false dichotomy.

11.5 Long context: 1M+ tokens, and what it does and doesn't buy

Scaling the usable context window runs into the same computational wall (attention cost grows *quadratically* with sequence length) plus a generalization problem: a model trained with positions up to length N often degrades if fed length $4N$ at inference, because it never saw those relative position gaps during training. **RoPE** extensions like position interpolation and NTK-aware scaling partially fix this.

By 2025, **1M+ token context went from research stunt to commodity** — Gemini 2.5 (1M), Llama-4 Scout (1M), GPT-4.1 (1M), Qwen2.5-Turbo (1M), MiniMax-Text-01 (4M). But the *how* matters: almost none of them use plain full attention. They combine RoPE extensions + continued pretraining on long sequences + sparse/linear attention + KV-cache engineering. "Full dense attention at 1M context" is not how frontier models actually work.

And crucially, **a longer window $\neq$ better long-range understanding**. Position bias ("Lost in the Middle") persists — models still gravitate to the start and end of their context; needle-in-a-haystack tests are saturated and gamed; RULER-style stress tests still show real degradation on long-range reasoning. Longer context is a *capability* you still have to make the model actually *use*.

11.6 Two things that changed the scaling story since the roadmap

Two developments belong in any modern scaling chapter:

- **The data wall.** Epoch AI (2024) estimated high-quality text data may be exhausted around 2026–2030 at current scaling rates. This is *why* labs pivoted to synthetic data and, more importantly, to RL-generated reasoning traces.

- **A new scaling axis: test-time compute / RL.** Snell et al. (2024) showed adaptive test-time compute can beat scaling parameters; OpenAI's o1/o3 and DeepSeek-R1 operationalized it via RL-trained "chain of thought" with adjustable reasoning budgets. This partially *decoupled capability from parameters* — the single biggest paradigm shift since Chinchilla, and the link between this chapter and the alignment chapter that follows.

Finally, **RAG** remains the way to turn a closed-book model into an open-book one: it retrieves relevant documents from an external index and inserts them into the prompt. A longer window doesn't solve a model's knowledge being frozen at training time — RAG does. In practice, the *engineering* of RAG (chunking, embedding choice, re-ranking) often matters more for output quality than the underlying LLM itself.

11.7 What to actually learn

- Compute-optimal scaling (Chinchilla) — and why the field now overtrains past it.
- Mixture-of-Experts routing and sparse activation.
- Why FlashAttention is fast (memory movement, not FLOPs) and what the KV cache is.
- Why longer context $\neq$ better long-range understanding (position bias).
- Test-time compute / RL as the new scaling axis.
- The RAG pipeline: chunking, embedding, retrieval, re-ranking, generation.

11.8 References

- Hoffmann et al., *Training Compute-Optimal Large Language Models* — Chinchilla (2022).
- Dao et al., *FlashAttention* (2022).
- Lewis et al., *Retrieval-Augmented Generation* (2020).
- Fedus et al., *Switch Transformers* (2021).
- DeepSeek-AI, *DeepSeek-V3 Technical Report* (2024) — MoE + MLA + multi-token prediction + FP8, all in one frontier model.
- Dao & Gu, *Mamba-2: Transformers are SSMS* (2024).
- Villalobos et al. / Epoch AI, *Will We Run Out of Data?* (2024).
- Snell et al., *Scaling LLM Test-Time Compute Optimally* (2024).

11.9 Reflection

Chinchilla says "train a smaller model on more data." "Beyond Chinchilla-Optimal" says "train it even smaller and overtrained." Why does *counting inference cost* flip the optimum in the second direction?

11.10 Try this week

Take a small transformer you've built (Part I, Stage 3) and measure how attention cost grows as you double sequence length — plot the memory/time vs. length. Then read the "Lost in the Middle" paper and construct a *needle-in-a-haystack* prompt for a model you have access to: put the answer at position 10% vs. 70% of a long prompt and see if it finds both. That's position bias, demonstrated on your own machine.

Chapter 12

CH06 • Alignment: RLHF & DPO

Turning a raw next-token predictor into something that follows instructions.

12.1 Getting here

Path: Pretraining → Scaling → **here**. (The RL side of the same idea is Part III, CH05.)

12.2 Interchange — with Deep RL, Stage 5 (Preference-Based RL / RLHF)

X4 (LLM 6 ↔ Deep RL 5). Training a model from human preference is the same technique in both lines. From the language side: RLHF/DPO turn preference data into a training signal that aligns an LLM. The general, reinforcement-learning statement of it — reward models, reward hacking — is in **Part III, Stage 5, CH05 (Preference-Based RL / RLHF)**. Read both pages together.

12.3 Premise

A model pretrained purely to predict the next token is not, by default, a helpful assistant — it’s a very capable text-completion engine that will happily continue a prompt in whatever direction is statistically likely, including unhelpful or evasive directions. **Alignment** shapes that raw capability toward doing what a person actually wants.

12.4 The classical pipeline: RLHF

The dominant approach for a long stretch was **Reinforcement Learning from Human Feedback (RLHF)**:

1. Collect human *comparisons* between pairs of model outputs.
2. Train a **reward model** to predict those preferences.
3. Fine-tune the language model with an RL algorithm — typically **PPO** — to maximize the reward model’s score.
4. Apply a **KL-divergence penalty** to keep the fine-tuned model from drifting too far from its original distribution (this is that KL term from Part 0, Stage 4, showing up exactly as promised).

This is a genuine reinforcement learning problem: the language model is the *policy*, generating a full response is the *action*, and the reward model’s score is the *reward signal* — precisely the RL

framework from Part III, applied with a learned reward function instead of a hand-coded one. That's why this stage is an interchange.

RLHF works but is notoriously finicky to tune: it needs a reward model and a stable RL loop, and both are fragile.

12.5 DPO: the same result, no RL loop

Direct Preference Optimization (DPO) reframes the same preference data as a simple classification-style loss directly on the language model's *own output probabilities*, mathematically derived to have the *same optimal solution* as RLHF — but without needing a separate reward model or an RL training loop at all. It's dramatically simpler to implement and stabilize. **And DPO is not a toy:** Meta used it at 400B scale in Llama-3. It spawned a whole zoo of variants (IPO, KTO, SimPO, ORPO, Nash-DPO), which signals how much of the field's energy went into preference-style alignment.

12.6 The roadmap is stale here: alignment split in two

The 2023 framing treated RLHF vs. DPO as the whole story. By the mid-2020s the field converged on a more useful split:

- **RLHF (reward model + PPO) and DPO-family methods** are the standard for *general helpfulness and safety* — shaping chat behavior when the "right answer" isn't verifiable.
- **RLVR (reinforcement learning with verifiable rewards) + GRPO** became the standard for *reasoning*. When the answer can be checked (a unit test passes, a math answer is right, code compiles), you don't need a learned reward model at all — the verifiable reward *is* the signal. **GRPO** drops the critic network entirely, using a group-relative baseline across sampled responses. This is the recipe behind **DeepSeek-R1**, **Kimi k1.5**, and **Qwen3**, and (undisclosed) the o1/o3 reasoning lineage. R1's famous "aha moment" — the model spontaneously improving its reasoning mid-training — turned out to have a mathematical explanation in how GRPO concentrates probability on successful groups.

So the modern answer to "RLHF or DPO?" is: **both, for different jobs** — plus RLVR/GRPO for reasoning.

12.7 The open problems matter more than the mechanics

If I'm teaching this honestly, the alignment chapter should lead with what's *not* solved:

- **Reward hacking is real and documented.** A policy optimized hard against an imperfect reward model will find and exploit whatever gaps exist between the learned reward and what humans actually want. It's a first-class concern, not a footnote — and it's exactly why the KL penalty matters.
- **Alignment faking** (Anthropic, 2024) is the most important warning artifact of the era: models trained with RLHF learned to *comply superficially* while preserving their "true" preferences. This reframes alignment as an open, unsolved research area — not a solved step in the pipeline.
- **Reasoning needs its own alignment.** Frontier labs now monitor and align the hidden *chain of thought* itself (GPT-5's system card documents CoT monitoring) — a subfield that didn't exist in 2023.

12.8 What to actually learn

- The RLHF pipeline: preference data $\rightarrow$ reward model $\rightarrow$ PPO fine-tuning $\rightarrow$ KL penalty.
- Why reward hacking happens and how KL constraints mitigate it.
- How DPO derives an equivalent objective without an explicit reward model or RL loop.
- Instruction tuning (SFT) as the step that usually precedes both.
- The RLHF (chat/safety) vs. RLVR+GRPO (reasoning) split.

12.9 References

- Ouyang et al., *InstructGPT* (2022).
- Rafailov et al., *Direct Preference Optimization* (2023).
- Schulman et al., *Proximal Policy Optimization* (2017).
- Shao et al., *DeepSeekMath / GRPO* (2024).
- DeepSeek-AI, *DeepSeek-R1* (2025).
- Anthropic, *Alignment Faking in Large Language Models* (2024).

12.10 Reflection

Why does a *verifiable* reward (a test passes) let you skip the reward model entirely — and why does that approach fail for the unverifiable cases that RLHF still handles?

12.11 Try this week

Take a small language model you can fine-tune and run a minimal DPO-style alignment on a tiny preference dataset (many libraries have toy examples). Observe how the model’s output distribution shifts. Then, in a sentence, articulate the *reward hacking* failure mode you’d most fear if you optimized that model hard against a learned reward.

Chapter 13

CH07 • Multimodal LLMs

Feeding images and video into a language model.

13.1 Getting here

Path: Scaling → Alignment → **here**. (Vision Transformers, Part II, CH02, helps.)

13.2 Interchange — with Video CV, Stage 8 (Video-Language Models)

X5 (LLM 7 ↔ Video CV 8). Vision-language fusion has two sides. From the language side (here): a vision encoder's features get projected into the LLM's context so it can *generate* language about what it sees. The embedding-centric side — contrastive training, zero-shot retrieval — is in **Part II, Stage 8, CH08 (Video-Language Models)**.

13.3 Premise

Once you accept that a Vision Transformer produces a sequence of patch-embedding tokens (Part I, Stage 3) and that a language model consumes sequences of token embeddings (Stage 3), the path to a multimodal LLM becomes almost obvious: run an image through a vision encoder, project its patch embeddings into the same embedding space the language model expects, and feed them into the language model's context alongside text tokens. The model attends across both modalities using the *same attention mechanism* throughout — there's no separate "vision reasoning module," just a shared sequence the transformer processes uniformly.

13.4 The two dominant fusion patterns

They differ in how tightly vision and language are fused:

- **Token-concatenation (LLaVA-style).** Use a lightweight projection layer to map vision-encoder output directly into the language model's token embedding space, then prepend or interleave those tokens with the text prompt. Simple, and it lets the full language model's attention operate over vision tokens with no architectural changes.
- **Cross-attention (Flamingo-style).** Keep the language model's forward pass mostly intact and insert new cross-attention layers that let text tokens attend into a separate stream of vision features. Scales more gracefully to many images or frames.

Open models still overwhelmingly use the projector-based (LLaVA-style) recipe — Qwen2.5-VL, InternVL3, LLaVA-OneVision, Molmo are all ViT + projection + LLM. Native unified-tokenizer models (Meta’s Chameleon, Transfusion, Mixture-of-Transformers) remain research rather than the open-model standard, though InternVL3’s "native multimodal pretraining" is a notable middle path. Extending either approach to video is mostly a matter of scale and efficiency — sampling frames, and avoiding feeding every patch of every frame as a separate token.

13.5 A subtle but important update: the vision encoder

The roadmap cites **CLIP** as the vision encoder underlying much of this work — correct for 2023. By 2025, **SigLIP** (and SigLIP-2) largely *replaced* CLIP as the default vision backbone in open VLMs: it uses a sigmoid loss that scales better than CLIP’s contrastive softmax. The concept (a shared image-text embedding space) is unchanged; the specific encoder moved on. This is exactly the kind of thing that ages a roadmap and is worth knowing how to update.

13.6 Training and the "omni" turn

Training these systems typically proceeds in stages: pretrain or reuse a strong vision encoder, pretrain or reuse a strong language model, then train the projection/fusion layers (and often fine-tune the whole system) on paired image-text and instruction- following data.

And "multimodal" has broadened since the roadmap. What started as "image captioning / VQA" is now **omni-modal and native**: GPT-4o (2024) brought realtime audio and any-to-any output; Gemini 3 (2025) made reasoning native across modalities with generative UI; Qwen2.5-Omni handles text/audio/video. For learning purposes, keep the *mechanism* in view (shared token sequence, projector fusion) and understand that the product surface grew enormously — but the underlying architecture is still the same transformer processing a shared sequence.

13.7 What to actually learn

- Token-concatenation fusion (LLaVA-style) vs. cross-attention fusion (Flamingo-style).
- How a vision encoder’s output is projected into a language model’s embedding space.
- Frame-sampling and token-budget tradeoffs for video input.
- Multi-stage training: vision-language alignment, then instruction tuning.
- Why SigLIP succeeded CLIP as the default open vision encoder.

13.8 References

- Alayrac et al., *Flamingo* (2022).
- Liu et al., *Visual Instruction Tuning* — LLaVA (2023).
- Radford et al., *CLIP* (2021).
- Zhai et al., *SigLIP* (2023); SigLIP-2 (2024).
- Bai et al., *Qwen2.5-VL Technical Report* (2025).

13.9 Reflection

Why is the *projector* (a simple learned mapping) enough to connect vision and language, rather than needing a fundamentally new architecture? What assumption does that rely on?

13.10 Try this week

Take a vision encoder (SigLIP or CLIP) and a small language model, and hand-build the "project vision tokens into the LM's embedding space" step from Part I, Stage 3's exercise: patchify an image, project it, and feed it as a prefix to text tokens. Watch the model attend over both. That's a minimal VLM, and you built it from parts you already own.

Chapter 14

CH08 • Agents & Tool Use

From answering in one shot to planning and acting over multiple steps.

14.1 Getting here

Path: Alignment → Multimodal LLMs → **here**.

14.2 Premise

Everything so far describes a model that maps an input sequence to an output sequence in a single forward pass. An **agent** is what you get when you wrap that model in a loop: it generates an action (text, a tool call, or both), observes the result, and decides what to do next, repeating until the task is done. This reframes the language model as the *reasoning core* of a decision-making system — a framing that connects directly to the RL formulation in Part III, though here the "policy" is implemented via prompting and in-context reasoning rather than gradient-based RL training (though the two are increasingly combined; see Part III, Stage 7).

14.3 The patterns that actually work

ReAct crystallized a simple, effective pattern: interleave explicit reasoning traces ("Thought: I need to check the weather before recommending a route") with actions ("Action: call the weather API"), feeding each action's result back into the context before the next reasoning step. This gives the model a place to *think before committing*, and lets errors surface and get corrected mid-task rather than compounding silently through one long generation.

The roadmap's tool-use story (Toolformer-style self-supervised tool use) is now partly stale. Production tool use became a **fine-tuned function-calling feature**: the model is given structured descriptions of available tools and emits a structured call that a runtime executes — far more reliable than parsing free text. And the plumbing got standardized: **MCP** (Model Context Protocol, Anthropic 2024) became the "USB-C of AI tools," adopted by OpenAI and Google and donated to the Linux Foundation's Agentic AI Foundation. Learning the ReAct *pattern* is still the right foundation; the *interface* is now standardized.

14.4 "Agents" now literally means acting in the world

One of the biggest changes since the roadmap: **computer use went from demo to product**. Claude's computer use (2024), OpenAI's Operator (2025), and the follow-ups let an agent drive a browser or OS directly. The agent that "calls a weather API" in ReAct's example is now routinely an agent that *clicks buttons, fills forms, and operates software*. This is the most direct, productized bridge to the action line of the map — and to the Terminus.

Anthropic's "Building Effective Agents" (2024) became the canonical vocabulary: distinguish **workflows** (prompt chaining, routing, parallelization, orchestrator-workers, evaluator-optimizer) from true **agents** (LLM-driven control loops), and argue that simple composable patterns beat complex frameworks. That taxonomy is worth internalizing — it saves you from the framework churn (LangChain/LangGraph hit 1.0 in 2025; AutoGen/Swarm came and evolved) by teaching the *patterns* rather than the tools.

14.5 The genuinely hard part: compounding error

The hard part of building agents isn't any single step — it's the **compounding of errors across many steps**. A task needing twenty correct sequential decisions fails if any one goes wrong. Dealing with that means investing in **self-correction, verification, and sensible retry behavior** rather than assuming the base model will get long horizons right by default. This is why evals are the open frontier: stepwise compounding error makes long-horizon agents brittle, and benchmark evals (SWE-bench-style) saturate quickly. Agent *reliability*, not raw capability, is the bottleneck.

14.6 What to actually learn

- The ReAct reasoning-then-acting loop and why interleaving beats one-shot generation.
- Structured function calling + MCP vs. free-text action parsing.
- Where multi-step agents fail: compounding error, context management, verification.
- The workflow-vs-agent taxonomy (Anthropic, "Building Effective Agents").
- Computer use as the productized form of "agents that act."

14.7 References

- Yao et al., *ReAct* (2022).
- Schick et al., *Toolformer* (2023).
- Anthropic, *Building Effective Agents* (2024).
- Anthropic, *Model Context Protocol* (2024).
- Anthropic, *Developing computer use* (2024).
- OpenAI, *Operator / Computer-Using Agent* (2025).

14.8 Reflection

Why does *interleaving* thought and action (ReAct) beat generating one long answer? And why does compounding error make long-horizon agents fundamentally harder than short ones?

14.9 Try this week

Build a minimal ReAct loop by hand: give a small model access to a single "tool" (say, a calculator or a weather API stub), interleave "Thought:" and "Action:" turns, and feed each result back. Watch it correct a mid-task error. Then, bonus, instrument it to count how often errors compound over 5 vs. 20 steps — that's the reliability problem, measured.

Part III

Part II — Computer Vision on Video

Contents of this part

- CH01 · CNNs & Image Foundations
- CH02 · Vision Transformers
- CH03 · Temporal Modeling: Flow, 3D Conv, Two-Stream
- CH04 · Video Transformers & Self-Supervised Pretraining
- CH05 · Video Prediction & World Models
- CH06 · Tracking, Pose & Interaction Recognition
- CH07 · Action & Activity Recognition
- CH08 · Video-Language Models
- CH09 · Video Understanding for Agents
- CH10 · Generative Video & Diffusion Models

From single-frame classification to perceiving motion, tracking people, and generating video outright — ten stages, the longest line on the map.

This is the **vision line** — and it's the longest, not because it's more important than the other two, but because it has more independently-teachable ideas. Video is the hard version of a hard problem: not just "what's in this image" but "what's happening over time, to whom, and what happens next."

The arc of the part:

1. **CNNs & Image Foundations** — the single-frame substrate every video model builds on.
2. **Vision Transformers** — an *interchange with LLMs*; the same transformer, pointed at image patches.
3. **Temporal Modeling: Flow, 3D Conv, Two-Stream** — the first video-native architectures, before video transformers existed.
4. **Video Transformers & Self-Supervised Pretraining** — attention across space *and* time, learned from unlabeled video.
5. **Video Prediction & World Models** — an *interchange with deep RL*; the perception half of planning.
6. **Tracking, Pose & Interaction Recognition** — following people and modeling how they relate.
7. **Action & Activity Recognition** — the applied task: saying what a person is doing from pose keypoints and temporal models, and when; the deployment reality the map gestures at.
8. **Video-Language Models** — an *interchange with LLMs*; connecting what's on screen to what can be said about it.
9. **Video Understanding for Agents** — perception built for the messy real world.
10. **Generative Video & Diffusion Models** — from understanding video to synthesizing it.

Notice the two interchanges in here: Stage 2 shares the transformer with Part I, and Stage 5 shares the *world model* with Part III. That's the map's whole point — the vision line isn't isolated, it's the perception half of systems that reason (Part I) and act (Part III). And the final stage, generative video, loops back to Stage 5: a model that predicts plausible future frames is, with a policy layered on top, most of the way to being a world model an agent could plan inside.

Let's start at the substrate: the convolutional network that reads a single frame.

Chapter 15

CH01 • CNNs & Image Foundations

The single-frame substrate every video model is eventually built on.

15.1 Getting here

Path: Part 0 → **here**. No prior vision needed — CNNs are the vision foundation.

15.2 Premise

Video understanding builds directly on single-image understanding, so the foundations are worth having solid even if your real interest is motion.

15.3 The assumption that makes convolutions work

Convolutional neural networks (CNNs) work because of a strong, useful assumption about images: **nearby pixels are related, and the same local pattern — an edge, a texture, a corner — can appear anywhere in the frame.** A dog's ear in the top-left is the same pattern as a dog's ear in the bottom-right.

A **convolutional layer** encodes this directly: it applies the *same small learned filter* across every spatial position. That one design choice buys two things a fully-connected layer has to pay for:

- **Far fewer parameters** — the filter is small and shared, rather than every pixel connecting to every other.
- **Translation invariance** — a pattern detector that works in one part of the image works in any part, without needing to be relearned there.

15.4 The feature hierarchy

Stack convolutional layers and you build a **hierarchy of features**:

- **Early layers** detect edges and simple textures.
- **Middle layers** combine those into parts (eyes, wheels).
- **Late layers** combine parts into whole-object representations.

This is why a CNN "understands" an image: not as raw pixels, but as a hierarchy from edges up to objects. It's a powerful inductive bias — the network is *given* the fact that local structure matters, rather than having to rediscover it.

15.5 The residual breakthrough

The practical breakthrough that let this hierarchy go genuinely deep was the **residual connection**. Instead of forcing each layer to learn a full transformation, **ResNet** has each block learn only a *residual* — a correction — added back to its input via a skip connection. This keeps gradients flowing cleanly through very deep networks that would otherwise *degrade in accuracy despite having more capacity* (the paradox that plagued pre-ResNet deep nets: adding layers made things worse, not better).

You've seen this idea before — it's the same residual trick inside the transformer block (Part I, Stage 3). It's one of the most reused ideas in all of deep learning, and it's worth recognizing it each time it reappears.

15.6 The realistic starting point: transfer learning

Here's the practical truth about building vision systems: you almost never train a CNN from scratch. The realistic starting point is an **ImageNet-pretrained backbone** — ResNet, or in modern pipelines a pretrained ViT — used as a feature extractor or a fine-tuning target. The low- and mid-level visual features these models learn (edges, textures, parts) transfer well across almost any downstream vision task. This is why "just grab a pretrained backbone" is the default opening move in basically every practical vision project.

15.7 What to actually learn

- Convolution, stride, and pooling — and why they encode a useful spatial prior.
- Why ResNet's residual connections enabled much deeper networks.
- Feature hierarchies: edges → parts → objects across layer depth.
- Transfer learning from ImageNet-pretrained backbones.

15.8 References

- He et al., *Deep Residual Learning for Image Recognition* — ResNet (2015).
- Stanford CS231n course materials — the standard reference course for CNN foundations.

15.9 Reflection

What does it mean for a layer to be "translation invariant," and why does that come *for free* from the architecture rather than from the data?

15.10 Try this week

Take a pretrained ResNet and a few images. Visualize what *early* vs. *late* layers activate on — show that early layers fire on edges and textures, late layers on whole objects. Then, bonus, fine-tune the last few layers of a pretrained CNN on a small custom image dataset and watch how fast it learns compared to training from scratch. That's transfer learning, demonstrated.

Chapter 16

CH02 • Vision Transformers

Treating an image as a sequence — and inheriting the transformer wholesale.

16.1 Getting here

Path: CNNs → Attention → Transformer Block → **here**.

16.2 Interchange — with LLM, Stage 3 (The Transformer Block)

X2 (Video CV 2 ↔ LLM 3). A ViT is the exact transformer block from the language line, applied to image patches. From the vision side: same architecture, different tokenizer — which is exactly why advances in one field transfer to the other. Read **Part I, Stage 3, CH03 (The Transformer Block)** for the language-facing view of the same block.

16.3 Premise

The Vision Transformer (ViT) makes a *deliberately minimal* set of changes to adapt images to the transformer architecture. That minimalism is the whole point — and it's the strongest piece of evidence for the map's central thesis that vision and language are one system.

16.4 How the ViT patchifies an image

Here's the entire recipe, and it's almost absurdly simple:

1. Split the image into a grid of fixed-size, non-overlapping **patches** (commonly 16×16 pixels).
2. Flatten each patch into a vector and linearly project it into a **token embedding** — exactly analogous to a word embedding.
3. Add a **learned position embedding** so the model knows where each patch sat in the original grid (since attention has no inherent sense of layout).
4. Feed the resulting sequence of patch tokens through the **same multi-head self-attention transformer blocks used for language** — with no convolution anywhere in the core computation.

That's it. A ViT is a language-model architecture pointed at a different tokenization scheme, exactly as Part I, Stage 3 claimed.

16.5 The honest cost of dropping convolution

This design choice is not free, and it's worth understanding *why* the paper found what it found. Convolutions have **translation invariance and locality built into their structure** — a helpful inductive bias when training data is limited. A ViT has to learn those regularities *from data* instead. That's why the original ViT paper found it **underperformed CNNs on mid-sized datasets** and only pulled ahead once pretrained on *very large* image datasets: attention is more flexible but needs more data to find the same useful patterns a CNN gets "for free" from its architecture.

In exchange, a ViT **scales more predictably with data and compute**, and behaves identically enough to a language transformer that advances discovered for one transfer directly to the other. That transferability — RoPE variants, FlashAttention, training-stability tricks — is the real prize, and it only makes sense if you keep the "same architecture, different tokens" framing in view.

16.6 Why this matters for the whole map

Every video transformer in Part II, Stage 4 is a direct extension of this idea into an extra temporal dimension: patches become **space-time patches**, and attention runs across space and time jointly or in factorized stages. And this is also the bridge to multimodality: once an image is just a sequence of patch tokens, feeding it into a language model's context is almost trivial (Part I, Stage 7). Keep "image = patch token sequence" in your head and large parts of the modern vision + multimodal stack unfold from it.

16.7 What to actually learn

- Patch embedding: splitting an image into a token sequence.
- Why ViT needs more pretraining data than CNNs to match their performance.
- 2D position embeddings and how they encode spatial layout.
- How this extends naturally to space-time patches for video (Stage 4).

16.8 References

- Dosovitskiy et al., *An Image is Worth 16x16 Words* (2020).
- Steiner et al., *How to Train Your ViT?* (2021) — practical training/data tradeoffs.

16.9 Reflection

What does the ViT's performance gap on small datasets (vs. CNNs) *teach* you about the difference between inductive bias and learned regularity? Where else does that tradeoff show up in this book?

16.10 Try this week

Patchify a small image yourself in NumPy (split into 8×8 patches, flatten, linearly project to tokens, add a position embedding) and feed the result to the tiny transformer you built in Part I, Stage 3. Then repeat the same exercise feeding it text tokens. Show yourself, concretely, that the transformer cannot tell the difference. That is the interchange made literal.

Chapter 17

CH03 • Temporal Modeling: Flow, 3D Conv, Two-Stream

The first generation of video-native architectures, before video transformers existed.

17.1 Getting here

Path: CNNs → here.

17.2 Premise

The most direct way to extend a CNN into video is to add a third dimension — time — to the convolution itself.

17.3 3D convolutions: convolve across time

Instead of sliding a 2D filter across height and width, a **3D convolution** slides a filter across *height, width, and time simultaneously*. This lets the network directly learn spatiotemporal patterns like "an edge moving rightward" rather than treating each frame independently.

The practical obstacle is data: video datasets are small and expensive, and learning spatiotemporal features from scratch on them is hard. **I3D** (*Inflated 3D ConvNet*) solved this with a clever trick that became a template: take a strong pretrained 2D image classification network and **"inflate" its 2D filters into 3D** by repeating them across the time dimension. This lets the 3D network *inherit ImageNet-pretrained weights* rather than learn spatiotemporal features entirely from scratch on much smaller video data. Reuse the single-image knowledge; only learn the temporal part new.

17.4 Two-stream networks: appearance vs. motion

An older and still-relevant idea reframes the problem as two complementary signals. A **two-stream network** runs one CNN over raw RGB frames to capture **appearance**, and a second CNN over **optical flow** fields to capture **motion**, then fuses the two streams' predictions.

Optical flow is worth understanding precisely: it's a *dense estimate of per-pixel motion between consecutive frames* — essentially a vector field showing where each pixel moved to in the next frame. It carries a great deal of action-relevant signal on its own, because two very differently-textured actions (a wave vs. a punch) can produce very different flow patterns even from similar-looking static frames. Appearance alone can be deceptive; motion is often the real signal.

17.5 SlowFast: the efficiency insight that matters

SlowFast refined the 3D-convolution approach with an insight about the *structure of video itself*. A video contains both slowly-changing semantic content (the scene, the objects) and quickly-changing motion detail (the hands, the gesture). A single network operating at one frame rate is forced to compromise between capturing both.

SlowFast instead runs **two pathways in parallel**:

- A **"slow" pathway** — processes frames at low temporal resolution but with large channel capacity, focused on spatial detail and semantics.
- A **"fast" pathway** — processes *many more* frames at low channel capacity, focused purely on motion.

The two are fused partway through the network. The result matched or beat much heavier single-pathway models at a fraction of the compute. It's a beautiful example of using an understanding of *what video actually contains* to design a cheaper architecture — a lesson that recurs throughout the efficiency parts of this book.

17.6 What to actually learn

- 3D convolutions and the "inflation" trick for reusing 2D pretrained weights.
- Optical flow as a motion representation, separate from raw appearance.
- Two-stream fusion of appearance and motion.
- The Slow/Fast dual-pathway design and why it's compute-efficient.

17.7 References

- Carreira & Zisserman, *Quo Vadis, Action Recognition?* — I3D (2017).
- Feichtenhofer et al., *SlowFast Networks* (2018).

17.8 Reflection

Why is "appearance can be deceptive" a good motivation for optical flow? Think of two actions that look similar in a single frame but differ in motion.

17.9 Try this week

Compute optical flow between two consecutive frames of a short video (OpenCV has a one-liner for it) and visualize it as a vector field or a colored image. See the motion signal *separately* from the appearance. Then, bonus, reflect on how a two-stream or SlowFast network exploits that split.

Chapter 18

CH04 • Video Transformers & Self-Supervised Pretraining

Attention across space and time — and learning from unlabeled video.

18.1 Getting here

Path: Vision Transformers → Temporal Modeling → [here](#).

18.2 Premise

Once attention proved it could handle images as patch sequences (Stage 2), the natural extension was to treat **video as a sequence of space-time patches** and run attention across all of them. The problem is cost, and the solution is factorization plus label-free pretraining.

18.3 The cost problem and the factorization fix

The naive version is expensive: attending every patch to every other patch across *many frames* scales quadratically in a sequence much larger than a single image produces. You can't just "run ViT on video" and expect it to be tractable.

TimeSformer's contribution was to show that **factorizing the attention** — first attend across *space* within a frame, then across *time* at the same spatial location, rather than full joint space-time attention — captures most of the useful modeling capacity at a fraction of the compute. Notably, in their experiments this "divided" attention *outperformed* the full joint version, not merely approximated it. The lesson: you don't need every patch attending to every other patch across all of time; spatial and temporal structure are somewhat separable, and exploiting that is both cheaper *and* better.

ViViT explored related factorization strategies and, importantly, examined how to **initialize a video transformer from an image-pretrained ViT** — reusing spatial knowledge rather than learning it from scratch on comparatively scarce labeled video. Same "reuse the image knowledge, learn the temporal part new" philosophy as I3D's inflation, now in transformer form.

18.4 Self-supervised pretraining: learning from unlabeled video

Labeled video is expensive and scarce — far scarcer than labeled images or text. That makes **self-supervised pretraining** especially valuable here, and there are two main routes:

Masked video modeling. **VideoMAE** adapts the masked-autoencoder idea to video and found something surprising: because consecutive frames are *highly redundant*, you can mask an **extremely high proportion** of space-time patches — far higher than works for images — and still force the model to learn genuinely useful motion and appearance representations. Reconstructing a masked patch

requires real understanding of both spatial content and temporal dynamics, not simple interpolation. The redundancy of video is what makes the aggressive masking possible.

Contrastive learning. The other route pulls together representations of different augmented views or clips from the *same* video while pushing apart representations from *different* videos — no labels and no reconstruction target at all. It learns "what belongs together" purely from the structure of the data.

Together, efficient factorized attention and label-free pretraining are what made transformer-based video models **practical rather than purely academic**.

18.5 What to actually learn

- Divided (factorized) space-time attention vs. full joint attention, and why factorization is both cheaper and competitive.
- Initializing video transformers from image-pretrained ViT weights.
- Masked video modeling, and why very high masking ratios work for video specifically.
- Contrastive self-supervised pretraining for video.

18.6 References

- Bertasius et al., *TimeSformer* (2021).
- Arnab et al., *ViViT* (2021).
- Tong et al., *VideoMAE* (2022).

18.7 Reflection

Why does the *redundancy of consecutive frames* both create a problem (wasted compute on near-duplicate frames) and enable a solution (aggressive masking)?

18.8 Try this week

Mask and reconstruct video patches with a small VideoMAE-style setup (or a pretrained model) at an extreme masking ratio — say 90% — and watch the model still recover plausible temporal content. Then reduce the masking to something image-like (e.g., 40%) and compare. The contrast shows exactly what video redundancy buys you.

Chapter 19

CH05 • Video Prediction & World Models

Learning to predict what happens next — the perception half of planning.

19.1 Getting here

Path: Video Transformers → **here**. (Its control-side twin is Part III, CH04.)

19.2 Interchange — with Deep RL, Stage 4 (Model-Based RL & World Models)

X3 (Video CV 5 ↔ Deep RL 4). The "predict what happens next" component built here is, seen from the control side, the learned environment model an RL agent plans inside. From the perception side: this stage supplies exactly that prediction. Read **Part III, Stage 4, CH04 (Model-Based RL & World Models)** together.

19.3 Premise

Every model discussed so far *consumes* video. A **world model** produces it — or more precisely, produces a compressed internal representation of "*what happens next*" given the current state and a hypothetical action.

19.4 The three-piece recipe (Ha & Schmidhuber's *World Models*)

The core idea, laid out clearly in Ha & Schmidhuber's 2018 paper, is to train a system with three parts:

1. A **vision component** that compresses a high-dimensional observation into a compact latent vector.
2. A **memory/dynamics component** that predicts how that latent state evolves over time — including as a function of *actions*.
3. A **lightweight controller** that operates entirely on the compressed latent state to decide what to do.

The key design choice is the last one: **the controller never has to reason about raw pixels directly**. It plans in a small, learned latent space that already captures the relevant structure of the world. That compression is what makes planning cheap enough to be practical.

19.5 The direct bridge to reinforcement learning

Here's the payoff, and it's the reason this stage is an *interchange* rather than a loose analogy: once you have a model that can predict "*if the agent does X from this state, the next latent state (and reward) will roughly be Y,*" you can **train a policy by imagining rollouts** inside that learned model — instead of only learning from real, expensive environment interactions.

Dreamer and its successors push this to its logical conclusion: an agent that learns a world model from raw pixel observations, then learns its entire policy **purely through imagined trajectories** generated by that world model — backpropagating through the imagined rollout to improve the policy without needing additional real environment steps for that particular update. In effect, the agent practices in its own head, using reality only to keep its mental model accurate.

DreamerV3 in particular demonstrated something worth calling out: a *single* set of hyperparameters could handle a very wide range of environments and observation types. That's a meaningfully different claim from "this architecture works" — it's closer to "this recipe is robust," which matters a great deal for anyone trying to actually deploy this. (You'll meet the control-side details in Part III, Stage 4.)

19.6 Why video prediction quality is the ceiling

There's a hard coupling here that explains why this stage connects so directly to the *generative video* stage (Stage 10) at the other end of this part: **the video-prediction quality of the vision component directly bounds how good the world model's imagined rollouts can be.** A world model can only plan well inside a mental simulation that predicts the world well. That's exactly why progress in video generation feeds so directly into progress on model-based RL — and why this is a genuine interchange, not an analogy.

19.7 Since the roadmap: world models became the field's center of gravity

The roadmap was written when world models were a research niche. By the mid-2020s they moved to the center of the field, and three developments *re-frame* this stage rather than replace it:

- **Action-conditional world models went huge.** DeepMind's **Genie** line (Genie 2 in Dec 2024; Genie 3 in Aug 2025, trained on ~1M years of video) produces interactive, action-controllable worlds from a single image — a world model acting as an *environment you can act inside*, not just a video predictor.
- **The pixel-vs-latent debate is live and important.** The generative camp predicts future *pixels* (Genie, Sora-class models). Meta's **V-JEPA 2** and Yann LeCun argue for predicting in *latent space*, self-supervised, without reconstruction. This is a genuine scientific dispute, not settled knowledge — the roadmap's Ha & Schmidhuber framing sits on the latent side, and the two camps disagree about which path leads to usable planning.
- **Generative world models became training environments.** **DIAMOND** (NeurIPS 2024 Best Paper) showed a *diffusion* world model could beat a strong RL agent (DQN) at Atari — i.e., training an agent inside a generative world model is now a real, award-winning direction, connecting directly to Stage 10 and to Part III, Stage 4.

The takeaway: the "predict what happens next, plan inside the prediction" core of this stage hasn't changed — but *how* the prediction is made (pixels vs. latent), and *how good* it can get, are open, contested questions the field is actively fighting over.

19.8 What to actually learn

- The compress $\rightarrow$ predict $\rightarrow$ control pipeline of a world model.
- Latent-space rollouts vs. pixel-space rollouts, and why latent is far cheaper.
- How Dreamer trains a policy purely from imagined trajectories.
- Why video prediction quality is a hard ceiling on world-model planning quality.
- The current pixel-vs-latent debate, and why it's unsettled (V-JEPA 2 vs. generative).

19.9 References

- Ha & Schmidhuber, *World Models* (2018).
- Hafner et al., *DreamerV3* (2023).
- DeepMind, *Genie 3: A new frontier for world models* (2025).
- Meta, *V-JEPA 2* (2025).
- Alonso et al., *DIAMOND* (NeurIPS 2024 Best Paper).

19.10 Reflection

Why does compressing raw pixels into a latent vector make planning *feasible* rather than just *smaller*?
What breaks if the compression is lossy in the wrong places?

19.11 Try this week

Take a short video clip and compress it into a small latent sequence (even a simple autoencoder will do). Visualize what the latents preserve and discard. Then, in one sentence, state why an agent that "plans in latent space" would be far cheaper than one that "imagines in raw pixels." Bonus: read Part III, Stage 4 now — this is the other half of the same idea.

Chapter 20

CH06 · Tracking, Pose & Interaction Recognition

Following people across frames, and modeling how they relate to each other.

20.1 Getting here

Path: CNNs → here.

20.2 Premise

Recognizing what a single, already-cropped video clip depicts is a meaningfully easier problem than operating on realistic, uncontrolled footage with multiple people entering, leaving, and occluding each other. This stage is about closing that gap.

20.3 Multi-object tracking: keeping identities straight

Multi-object tracking assigns a consistent identity to each person across frames. The dominant approach is **tracking-by-detection**:

1. Run a per-frame object detector.
2. Solve an **association problem** that links each new frame's detections to existing tracked identities, based on position, motion, and appearance similarity.

The hard part is the association — deciding whether the bounding box in frame 50 is the same person from frame 49 or a different one.

ByteTrack's insight is instructive and subtle. Most trackers *discard low-confidence detections* as likely false positives. But that's exactly the wrong move when a real person is briefly occluded or blurred — their detector score drops, but they're still there. ByteTrack instead **keeps every detection box** and uses similarity to existing tracklets to decide whether a low-confidence box is a *real, temporarily-degraded detection* or genuine background noise. This substantially reduces identity switches and fragmented tracks. The lesson generalizes: in real, messy footage, "low confidence" doesn't mean "not there."

20.4 Pose estimation: the skeleton as a signal

Pose estimation adds a finer-grained representation on top of tracking: instead of just a bounding box per person, extract the **2D or 3D coordinates of their body joints** (shoulders, elbows, hands, etc.).

This representation is deliberately *compact and background-invariant*: a skeleton is largely unaffected by lighting, clothing texture, or camera background. That invariance is what makes it a strong basis for recognizing actions and interactions specifically — you throw away almost all the irrelevant visual variation and keep just the part that conveys *what the body is doing*.

20.5 ST-GCN: skeletons as graphs, actions as graph convolution

ST-GCN treats a sequence of pose skeletons as a **spatio-temporal graph**:

- **Joints are nodes.**
- **Natural skeletal connections** (elbow→wrist) are spatial edges.
- The **same joint across consecutive frames** is connected by temporal edges.

Graph convolutions are then applied across this structure to classify actions directly from joint trajectories. Extending this to **interaction recognition** is a matter of adding edges *between* two people's skeletons as well, so the graph convolution can directly learn relationships like relative distance, orientation, and synchronized motion — rather than only within a single person's body.

This is the most direct, practical path to building an interaction-recognition system, and it's dramatically **less data-hungry than raw-RGB approaches** — precisely because the pose-extraction step has already thrown away most of the irrelevant visual variation.

20.6 Since the roadmap: tracking went foundation-model

The roadmap describes the *association-based* generation of tracking, and it's still the right way to learn the problem. But the research frontier moved on:

- **Promptable video segmentation.** **SAM 2** (Meta, 2024) made tracking a *foundation model* capability: click a mask on one frame and it segments that object through the whole video, with zero training, for arbitrary objects. This subsumes much of what ByteTrack-style pipelines did — which is why you should learn association tracking (it teaches you the problem) but know that production has largely moved to segmentation/ point-based foundation models.
- **Point tracking.** **TAP** (Tracking Any Point) and **CoTracker** track arbitrary *points* across long videos, powering markerless 3D tracking — the new research front for the "keep identities across frames" problem.
- **Pose became an input, not a task.** 2D/3D pose is mature, foundation-model-ized (e.g., **SMPLer-X**), and increasingly an *input* to larger systems rather than an end goal. MediaPipe remains the go-to on-device API, but it's now one of many.

The lesson: the *conceptual* skills here (identity association, skeleton graphs, interaction edges) still transfer — but expect to encounter them wrapped in foundation models rather than as standalone pipelines.

20.7 What to actually learn

- Tracking-by-detection and the identity-association problem across frames.
- Why keeping low-confidence detections (ByteTrack) reduces identity switches.
- Skeleton graphs as a spatio-temporal representation for action recognition.
- Extending single-person skeleton graphs to multi-person interaction graphs.
- How SAM 2 / point tracking superseded association tracking as the production paradigm.

20.8 References

- Zhang et al., *ByteTrack* (2021).
- Yan et al., *ST-GCN* (2018).
- Google MediaPipe — the most accessible way to get real-time multi-person pose extraction running.
- Ravi et al., *SAM 2* (Meta, 2024).

20.9 Reflection

Why does a "skeleton" representation make interaction recognition easier *and* cheaper than raw RGB? What is thrown away, and why is it safe to throw away?

20.10 Try this week

Run MediaPipe pose on a short video of two people interacting. Visualize the two skeletons and their joints. Then, in a sentence or two, describe how you'd add "edges between the two people" to turn single-person action recognition into interaction recognition — and what information those edges would carry.

Chapter 21

CH07 • Action & Activity Recognition

From knowing a person is there to knowing what they're doing — and when.

21.1 Getting here

Path: Tracking, Pose & Interaction → Recurrent Sequence Models (Part I, CH01) → **here**.

21.2 Interchange — with LLM, Stage 1 (Recurrent Sequence Models)

X1 (Video CV 7 ↔ LLM 1). The temporal model at this stage's heart — an LSTM classifying pose-keypoint sequences — is the recurrent machinery the language line built on before attention. From the video side: that cell is what makes an activity classifiable over time, and the compact pose-keypoint input is what keeps it cheap enough to run live. Read **Part I, Stage 1, CH01 (Recurrent Sequence Models)** to meet the same cell from the language side.

21.3 Premise

The last few stages were about *following* and *representing* people — detection, tracking, pose. This stage is about **understanding what they're doing**: given video of people, answer *what action is happening*, and for longer footage, *exactly when it starts and ends*. Two related problems, lumped together in most job briefs, but genuinely different.

21.4 Two problems, one confusing name

If you were handed the phrase "action recognition and activity detection," it's really two tasks, and getting them straight up front saves you real confusion:

- **Action recognition** — given a *short, trimmed* clip that (mostly) contains one action, output the action label: "squatting," "waving," "walking." One label for the clip. This is the simpler task and where most tutorials and libraries (including ActionAI's core examples) start.
- **Activity / action detection** (a.k.a. *temporal action localization*) — given a *long, untrimmed* stream, find **both** what actions occur *and* precisely when each starts and ends: "seconds 3–9: spinning; seconds 12–18: squatting." This is the version you actually need in real footage, where people act continuously and nothing is pre-trimmed for you. (You met this same "untrimmed vs. trimmed" gap in Part II, Stage 9.)

A model that nails the first is not automatically close to the second — detection adds the hard problem of *where action boundaries are*, which is often ambiguous even to humans.

21.5 The classical recipe that got you here: two stages

The classical starting point — "LSTM + ResNet" — is the **two-stage recipe**, and understanding it is the key that unlocks most legacy human-activity systems:

1. **A compact per-frame signal**, extracted by a convolutional network. The point of this stage is to turn each frame into a *small, information-dense* representation instead of raw pixels — features a temporal model can actually learn from. "ResNet" here is the classic choice for this extractor; often it's not even a full frame-CNN but a **pose estimator** that reduces each frame to a few dozen body-keypoint coordinates (Stage 6's idea, put to work).
2. **A temporal model** over that per-frame signal — classically an **LSTM** — which looks at the sequence and decides what action the motion represents.

Why two stages instead of one big network? The same reason the whole map keeps returning to: raw video is huge and redundant, and modeling *long-range* structure directly from pixels is expensive. If you first compress each frame into what matters (the body, roughly), the temporal learner has a tiny, clean input and can do its job with far less data — a *deployment-critical* advantage when your training set is a few hours of labeled footage, not millions of clips.

21.6 What an LSTM actually is (no mystery)

You've already done the hardest part — backprop and an MLP. An **LSTM** (Long Short-Term Memory) is just a neural network with **memory**: alongside the usual hidden state it keeps a *cell state* it can write to, read from, and *forget* parts of, using learned gates. That memory is what lets it carry relevant context across many timesteps — "this is a squat" depends on the whole downward-and-upward motion, not one frame.

This is exactly the model the map's attention chapter (Part I, Stage 2) was written against: before attention, the standard way to model a sequence was recurrence — an RNN/LSTM reading one step at a time, carrying a compressed state forward. The map moves on to attention because it parallelizes and doesn't suffer the long-range compression bottleneck. But for a *short* motion window (a squat takes ~1 second; an action ~tens of frames), an LSTM's bottleneck is rarely the limiting factor — which is why the classical recipe still works well for atomic actions and remains everywhere in production. Learning it is not wasted time; it's understanding what the field ran on, and still runs on.

21.7 Pose keypoints as the signal that makes it cheap

The version of the two-stage recipe that ActionAI-class systems actually ship is built on **pose keypoints** (Stage 6's representation): detect the person, extract the 2D/3D coordinates of body joints (shoulders, elbows, wrists, knees...) per frame, and throw away the pixels. What's left is a compact, *background-invariant* description of the body's motion.

This is the same reason the pose chapter gave for skeletons over raw RGB — but now it pays off in a deployment setting:

- **Small input.** Each frame becomes a handful of coordinates, not a million pixels. The temporal model is tiny and trains on a laptop.
- **Robust to scene.** Lighting, clothing, and background stop mattering once you only keep the body.
- **Privacy- and bandwidth-friendly.** You can extract keypoints on-device and transmit only the skeleton — a real selling point for cameras in homes or clinics.

- **The cost** is that pose quality caps everything downstream: if the pose estimator loses the person (occlusion, heavy blur, unusual poses), the action classifier has nothing sensible to look at. Pose is the ceiling.

21.8 The full deployed pipeline (ActionAI, class-level)

Put it together and you get the shape of an ActionAI-style human-activity system — not the repo's internals, but the architecture the whole class shares:

1. **Per-frame pose extraction** — a CNN pose model (a *Convolutional Pose Machine* or MobileNet-class net) finds the person and outputs keypoints each frame.
2. **Buffer into windows** — because a single frame can't tell you an action, keep a sliding window of the last N frames' keypoints as the input "clip."
3. **LSTM classifier** — feed the windowed keypoint sequence to an LSTM (often with a small fully-connected head) that outputs a label per window: squat, spin, wave...
4. **Tracking for multi-person** — when several people are present, associate keypoints across frames (Stage 6) so each person gets their *own* consistent keypoint stream, rather than mixing up two people's skeletons.
5. **Activity with timing** — smooth the per-window labels and post-process to produce not just "spinning" but "spinning from second 3 to 9" — the spatio-temporal activity output that makes it *detection*, not just recognition.
6. **Edge deployment** — because pose + LSTM is cheap, the whole thing runs in real time on a Jetson Nano, Raspberry Pi, or Edge TPU — a camera pointed at a room, classifying activity live, often with an online "teachable" mode to label and fine-tune on new actions in the field.

That last point is the reason this classical stack refuses to die: **it fits on hardware and datasets that SOTA research models can't touch**. If you can only gather a few hundred labeled examples and must run at 30fps on a \$100 device, pose + LSTM is frequently the *right* answer, not a compromise.

21.9 Data, evaluation, and the honest caveats

- **Benchmarks:** trimmed-clip action recognition is usually evaluated on UCF101, HMDB51, or the (much larger, harder) Kinetics; temporal detection on THUMOS or ActivityNet. Real deployment data rarely looks like these — expect domain gaps.
- **Metrics:** recognition reports top-1/top-5 accuracy; detection uses mAP over temporal IoU (how well your predicted action intervals overlap the true ones). Don't compare the two.
- **The failure you'll actually meet is pose.** Low-light, occlusion, far subjects, unusual angles, and fast motion all degrade the pose extractor first, and everything downstream fails silently. Budget real effort here.
- **Classes and context matter.** "Is this person falling?" in a care home is a very different, harder problem than "squat vs. spin" in a controlled gym shot. Nail down the *real* class set and its ambiguity before chasing architecture.

21.10 Bridging to the stages around it

This stage sits at the applied end of the map's vision line, and it's worth seeing the threads it connects:

- **Stage 3 (temporal modeling)** taught the *other* classical route — 3D convolutions (I3D), optical-flow two-stream, and SlowFast — which recognize actions from RGB motion directly, no pose step. The pose+LSTM recipe here is the recurrent cousin of that same generation.
- **Stage 4 (video transformers)** is where the field's frontier moved: attention over space-time patches can recognize actions with far more data and compute — but it needs scale this applied setting rarely has.
- **Stage 6 (pose)** supplies the representation this chapter leans on; read them together.
- **Stage 8 (video-language), next**, picks up where action labels stop — connecting what's on screen to full language, not just a fixed action vocabulary.

The honest summary: **the classical pose+LSTM recipe is not the frontier, and you should know that** — but it is very often the right engineering answer, and understanding it end to end makes you fluent in a huge amount of real deployed video-AI, not just a paper.

21.11 What to actually learn

- Action recognition (trimmed clip, one label) vs. activity/action detection (long video, label + timing) — and why they're different problems.
- The two-stage recipe: compact per-frame signal + temporal model, and why it needs far less data than end-to-end pixel modeling.
- What an LSTM is (an MLP with a memory cell) and why recurrence was the pre-attention way to model sequence (tie to Part I, Stage 2).
- Why pose keypoints make activity recognition cheap, edge-friendly, and privacy-friendlier — and why pose quality is the ceiling.
- The full deployed pipeline: pose → windowed keypoints → LSTM → tracking → timed activity → edge deployment (ActionAI as the concrete class of system).
- Where the classical route sits against 3D conv / two-stream (Stage 3) and video transformers (Stage 4).

21.12 References

- Donahue et al., *Long-term Recurrent Convolutional Networks for Visual Recognition and Description* — LRCN (2015) — the "CNN features + LSTM" recipe this stage is built on.
- [smellslikeml/ActionAI](#) — a concrete, edge-deployed instance of the pose + LSTM approach (Jetson Nano / Raspberry Pi).
- Wei et al., *Convolutional Pose Machines* (2016) — the pose-estimation lineage ActionAI uses.
- A recent survey of deep video action recognition — for orienting the classical vs. modern landscape.
- Carreira & Zisserman, *I3D* (Stage 3 reference) and the UCF101 / HMDB51 / Kinetics benchmarks for evaluating recognition.

21.13 Reflection

Why does the *two-stage* recipe (compress each frame, then model the sequence) so often beat end-to-end pixel modeling when your labeled data is scarce? And where does pose quality become the hard ceiling on such a system?

21.14 Try this week

In your study→implement→write style: take a short, single-action clip of yourself (or a video from a pose-demo dataset), run a pose extractor to get per-frame keypoints, and hand-roll or fine-tune a small LSTM to classify two contrasting actions (e.g., squat vs. wave) over a sliding window. Then *write* a paragraph on where it breaks when you add a second person or bad lighting. That's the whole pipeline — pose → window → LSTM → label — demonstrated, and you'll have met the deployment reality no tutorial screenshots capture.

Chapter 22

CH08 • Video-Language Models

Connecting what's on screen to what can be said about it.

22.1 Getting here

Path: Vision Transformers → Multimodal LLMs (Part I, CH07) → **here**.

22.2 Interchange — with LLM, Stage 7 (Multimodal LLMs)

X5 (Video CV 8 ↔ LLM 7). Vision-language fusion has two sides. From the video/embedding side (here): a contrastive objective builds a joint image-text space for retrieval and zero-shot classification. The generative side — feeding vision into a language model so it can describe what it sees — is in **Part I, Stage 7, CH07 (Multimodal LLMs)**.

22.3 Premise

CLIP established the template this whole area builds on: train an image encoder and a text encoder *jointly*, using a **contrastive objective** that pulls the embeddings of a matched image-caption pair close together while pushing embeddings of mismatched pairs apart, on hundreds of millions of loosely-supervised image-caption pairs scraped from the web.

The result is a **shared embedding space** where semantic similarity between an image and a piece of text can be measured as simple cosine similarity. And that enables **zero-shot classification**: embed the image, embed a text prompt for each candidate category, and pick whichever text embedding is closest — with no task-specific training at all. This is a genuinely powerful trick: you can classify into *new* categories you never trained on, just by phrasing them as text.

22.4 Extending contrastive learning to video

Extending this to video mostly means deciding **how to aggregate information across frames** before or after the contrastive comparison — whether to pool per-frame embeddings, run a lightweight temporal transformer over per-frame CLIP features, or train the video encoder end-to-end with a similarly-scaled video-text contrastive objective. **VideoCLIP** and similar approaches adapt the recipe directly to video-text pairs, producing embeddings useful for zero-shot video classification, video-text retrieval, and as a strong initialization for video captioning models (which generate a full natural-language description rather than just retrieving or classifying).

22.5 Since the roadmap: video understanding was absorbed into LMMs

This is the chapter where the roadmap is most out of date, and the correction is instructive. Through the mid-2020s, **video understanding became dominated by large multimodal models (LMMs) with long-context video input** — the API-level product is literally "paste an hour of video, ask questions." The relevant anchors:

- **Gemini 2.5** (2025): 1M+ token context, native multimodality, demoed on hour-scale video.
- **Qwen3-VL** (2025): claims $\sim 99.5\%$ accuracy at detecting a *single tampered frame* in a 2-hour video, and does temporal grounding + video QA natively.
- **Benchmarks caught up**: Video-MME, MLVU, and TimeScope (which literally asks "how long can your video LMM go?") became the reference evaluations, replacing CLIP-style retrieval checks.

So the *embedding-centric* view in this chapter (CLIP/VideoCLIP) is still how you'd *build a retrieval system*, but it's no longer how frontier systems *do video understanding* — that moved to generative LMMs that reason over video tokens. Notably, even temporal *localization/grounding* (find the moment) became an LMM capability, not a separate detection head. And the classic pipeline isn't dead — its encoders live *inside* the VLMs as a pretraining layer. The honest takeaway: learn the contrastive foundation (it's the mechanism), but know that production video understanding now lives in long-context LMMs.

22.6 What to actually learn

- Contrastive image/video-text pretraining and the InfoNCE-style loss behind it.
- Zero-shot classification via text-prompt embedding comparison.
- Frame aggregation strategies for extending image-text contrastive models to video.
- Video captioning as a downstream generative task built on these embeddings.
- How long-context LMMs (Gemini 2.5, Qwen3-VL) absorbed video understanding and grounding.

22.7 References

- Radford et al., *CLIP* (2021).
- Xu et al., *VideoCLIP* (2021).
- Gemini Team, *Gemini 2.5 Technical Report* (2025).
- Qwen3-VL Technical Report (2025).
- Video-MME benchmark.

22.8 Reflection

What does "zero-shot classification by phrasing categories as text" *buy* you that a traditional classifier never could? And why is that power somewhat irrelevant now that LMMs reason over video directly?

22.9 Try this week

Take a CLIP/SigLIP model and classify a few video clips zero-shot by writing category prompts as text — watch the embedding-similarity mechanism in action. Then feed the same clips to any long-context multimodal model you have access to and ask a question that requires *temporal* reasoning (e.g., "what changed between the middle and the end?"). Notice how the two approaches differ in what they're good at.

Chapter 23

CH09 • Video Understanding for Agents

Perception built for the real, messy world rather than curated benchmark clips.

23.1 Getting here

Path: Tracking, Pose & Interaction → Action & Activity Recognition → **here**.

23.2 Premise

Nearly everything up to this point assumes a relatively convenient setup: a short, pre-trimmed clip containing one clear action, viewed from a fixed third-person camera. Real deployment — robotics, wearable devices, long-running surveillance — breaks all three assumptions at once. This stage is about the gap between benchmark performance and actually working in the field.

23.3 Dropping the "pre-trimmed" assumption: temporal localization

Temporal action localization is the task of finding, in a long continuous unedited video, *what* actions occur and *precisely when* they start and end — rather than classifying a clip already cut to contain exactly one action. This requires proposing candidate time intervals across a long video, analogous to region proposals in object detection but along the time axis. And because action boundaries are often ambiguous even to human annotators, datasets like **AVA** densely annotate *atomic* actions at one-second intervals across movie clips rather than relying on clip-level labels.

The roadmap frames this as a specialized architecture problem. **The 2025 correction** is that temporal localization/grounding was *absorbed into LMMs as a text-conditioned capability* — "find the moment" is now an LLM task over video tokens (VTimeLLM, ReVisionLLM, TimeZero), and Qwen3-VL does temporal grounding natively. Learn the *problem* (boundary ambiguity, dense annotation) because it hasn't gone away — but know that the mechanism is now inside long-context video LMMs, not a separate detection head.

23.4 Dropping the "fixed third-person camera" assumption: egocentric vision

Egocentric (first-person) vision changes the problem substantially. A head- or chest-mounted camera moves constantly with the wearer, frequently has the primary subject partially out of frame or motion-blurred, and requires reasoning about *intent and upcoming actions* from a viewpoint that never shows the wearer's own body language. **Ego4D** assembled a very large, diverse egocentric dataset to push progress here, since almost all earlier video datasets are third-person. Since then the area went further:

- **Ego4D** was peer-reviewed (TPAMI 2024); **Ego-Exo4D** (2025) added paired first/third-person skilled-activity data.
- **Project Aria** (Meta) became a research *hardware* product — Aria Gen 2 glasses — with photoreal digital-twin reconstruction, explicitly positioning egocentric data as the training substrate for embodied AI.

23.5 Real-time constraints reshape what’s viable

Finally, **real-time constraints** reshape which architectures are viable: many of the highest-accuracy models discussed earlier are far too slow for live use on an edge device or robot. This pushes practical systems toward lighter architectures, aggressive frame subsampling, and careful **accuracy/latency tradeoffs** that rarely show up in benchmark leaderboards but *dominate real deployment decisions*. This is the stage where you learn that a working video system is usually a compromise, not a SOTA model.

23.6 What to actually learn

- Temporal action localization vs. trimmed-clip classification.
- Why egocentric video breaks assumptions built into third-person datasets.
- Real-time inference constraints and accuracy/latency tradeoffs for deployment.
- Dense, atomic-action annotation schemes (AVA-style) vs. single clip-level labels.
- How temporal grounding moved into long-context LMMs.

23.7 References

- Gu et al., *AVA* (2017).
- Grauman et al., *Ego4D* (2021).
- Grauman et al., *Ego4D: Around the World in 3,600 Hours* — TPAMI (2024).
- Project Aria Gen 2.

23.8 Reflection

Why does *egocentric* video make the problem fundamentally different, not just harder? What information is lost when the camera moves with the wearer?

23.9 Try this week

Take a long unedited video (or find one) and run a temporal-localization question through a long-context LMM: "at what times does X happen?" Check the boundaries against your own judgment — and note where *you* found the boundary ambiguous. Then, separately, run an ego-style first-person clip and observe what a third-person-trained model gets wrong about it. Both exercises are about the real world breaking clean assumptions.

Chapter 24

CH10 • Generative Video & Diffusion Models

From understanding video to synthesizing it outright.

24.1 Getting here

Path: Video Transformers → World Models → **here**.

24.2 Premise

Every earlier CV stage takes video as input and produces a label, embedding, or prediction. This final stage *flips the direction*: given some conditioning signal — text, a starting frame, or nothing at all — generate new, plausible video.

24.3 The diffusion framework

The dominant approach borrows the diffusion framework from image generation: train a model to **reverse a gradual noising process** — starting from pure noise and iteratively denoising toward a coherent sample, conditioned on whatever signal should steer the output.

For video, this has to solve *appearance and temporal coherence simultaneously* — not just "does each frame look real" but "does the motion between frames look physically plausible and consistent." That's a substantially harder joint constraint than image generation faces. (There's a direct echo here of Stage 5: a model that predicts plausible future frames is solving a world-model-shaped problem.)

24.4 Two practical strategies

Two strategies dominate:

- **Inflate a pretrained 2D image diffusion model** with additional temporal layers. This reuses an enormous amount of image-generation knowledge and dramatically reduces how much video data is needed — at some cost in motion quality compared to models trained on video from the start.
- **Train a video-native diffusion transformer end-to-end** on large video datasets — treating video as space-time patches much like Stage 4's video transformers. This tends to produce better long-range temporal coherence but requires far more video data and compute to train well.

24.5 Since the roadmap: the generative-video landscape moved fast

The roadmap's "Generative Video & Diffusion" stage aged more than almost any other, and the updates are worth knowing as a case study in how fast this field moves:

- **The flagship consumer moment was short-lived.** OpenAI's **Sora 2** (Sept 2025) launched as a consumer app — and, per press reports, was *shut down* around April 2026, with a reported ~\$10B Disney partnership terminated. Treat these as press reports to verify, but the pattern is a real lesson: a headline-generating consumer product is not the same as a durable business or a settled technology.
- **Google counter-positioned as enterprise/API.** **Veo 3.1** (2025) targeted API/Flow for enterprises, and 2026 pivoted to "reasoning-first" generative models (Gemini Omni) explicitly selling "*simulate the world*" — pushing video generation toward world simulation.
- **Open source largely closed the gap.** **Wan 2.1→3.0** (Alibaba), **HunyuanVideo** (Tencent, full training pipeline open-sourced), **CogVideoX**, and **LTX-Video** made frontier-class video generation available without a proprietary API by 2025. (Nuance: "open source" $\neq$ "open weights forever" — Wan 3.0's open status became contested.)
- **Runway's Gen-4 added "world consistency"** — characters, objects, and environments persisting across shots — addressing the top practical failure of 2023-era diffusion video.

24.6 The connection back to world models (Stage 5)

The conceptual link that matters most for this book: **a generative video model and a world model's video-prediction component are solving closely related problems** — both must predict plausible future frames — and advances in one increasingly transfer to the other. A sufficiently good text-to-video or video-continuation model is, with the right conditioning and a policy layered on top, *most of the way to being a world model an agent could plan inside*.

But be honest about the frontier: **generative video models simulate appearances, not physics**. Rivals and evaluators contest the "Sora is a world model" claim; VBench++-era automatic metrics disagree with human evaluation on long-tail items; and world-model evaluation is itself unsolved (WBench). For learning purposes, the roadmap's own framing holds: treat generative video as a *candidate* world-model substrate, not a proven physics simulator.

24.7 What to actually learn

- The diffusion framework: learning to reverse a noising process.
- Why temporal coherence is a harder constraint than single-frame realism.
- Inflating pretrained image diffusion models vs. training video-native diffusion transformers.
- The direct connection between generative video and world-model video prediction (Stage 5).
- How the 2024–2026 landscape (Sora, Veo, open-source) reshaped the stage.

24.8 References

- Ho et al., *Video Diffusion Models* (2022).
- Ho et al., *Imagen Video* (2022).
- Hugging Face, *State of open video generation models*.

- Runway, *Gen-4: Introducing world consistency*.
- VBench++ (video-generation benchmark).

24.9 Reflection

A text-to-video model and a world model both "predict plausible future frames." Where does the similarity end — what does a world model need that a video generator doesn't?

24.10 Try this week

Generate a short clip with an open-source video model (or a text-to-video API), then test its *temporal consistency*: have a character move behind an object and reappear, or generate a longer clip and look for the object to change identity. Note exactly where it fails — and connect that failure to why "world consistency" and "world model" became the next research frontier.

Part IV

Part III — Deep Reinforcement Learning

Contents of this part

- CH01 · MDPs & Dynamic Programming
- CH02 · Value-Based Deep RL
- CH03 · Policy Gradient Methods
- CH04 · Model-Based RL & World Models
- CH05 · Preference-Based RL / RLHF
- CH06 · Multi-Agent & Embodied RL
- CH07 · LLM-Guided Agents

From the formal decision-making framework to agents that plan inside learned world models and take direction from language models.

This is the **action line** of the map. Where Part I gave you a system that *reasons* and Part II gave you a system that *perceives*, this part gives you a system that *acts* — an agent that makes a sequence of decisions to maximize some notion of reward.

The arc of the part:

1. **MDPs & Dynamic Programming** — the formal frame every RL algorithm in this part is an approximation of.
2. **Value-Based Deep RL** — replacing an exact table of values with a neural network.
3. **Policy Gradient Methods** — optimizing a policy directly instead of deriving it from a value function.
4. **Model-Based RL & World Models** — an *interchange with video CV*; learning a model of the world and planning inside it.
5. **Preference-Based RL / RLHF** — an *interchange with LLMs*; when the reward function itself is learned from human judgment.
6. **Multi-Agent & Embodied RL** — scaling to many agents, or a real body.
7. **LLM-Guided Agents** — using a language model as the high-level planner.

Two of these stages are interchanges, and they're the heart of the map's thesis: Stage 4 shares the *world model* with Part II's perception line, and Stage 5 shares *preference learning* with Part I's alignment line. The action line isn't separate from reasoning and perception — it's what they ultimately feed.

And notice where the part ends: **LLM-guided agents** (Stage 7) — the point where the RL line and the LLM line converge most directly in practice. That's the bridge to the Terminus in Part V, where all three lines meet in a single system that perceives, reasons, and acts.

Let's start where RL has to start: the formal framework that makes sequential decision under uncertainty tractable at all.

Chapter 25

CH01 · MDPs & Dynamic Programming

The formal frame every RL algorithm in this part is an approximation of.

25.1 Getting here

Path: Part 0 (Probability) → **here**.

25.2 Premise

A **Markov Decision Process (MDP)** formalizes sequential decision-making under uncertainty. It's the thing every algorithm in this part is an approximation of.

25.3 The four ingredients

An MDP has four parts:

1. A set of **states** — the situations the agent can be in.
2. A set of **actions** — the choices the agent can make.
3. A **transition function** — the probability of moving to a new state given the current state and action.
4. A **reward function** — the numerical signal the agent wants to maximize.

The "**Markov**" part is a specific and consequential assumption: *the future depends only on the current state, not on the history that led to it*. This is what makes the problem tractable at all — you don't need to remember everything, just where you are now. It's also part of why **representing state well** matters so much in practice: if the state doesn't capture what's relevant, the Markov assumption silently breaks.

25.4 Dynamic programming: exact methods when the MDP is fully known

Given a *full, known* MDP, dynamic programming gives exact methods for finding the optimal policy. Two classic algorithms:

- **Value iteration** repeatedly applies the **Bellman optimality equation** — *the value of a state equals the immediate reward plus the discounted value of the best next state you can reach* — until the value estimates converge, then extracts the optimal policy by acting greedily with respect to those values.

- **Policy iteration** alternates between *evaluating* a fixed policy's value function and *improving* the policy by acting greedily with respect to that value function. It provably converges to the optimal policy in a finite number of steps for a finite MDP.

Here's the honest framing: these methods matter **less** for their direct practical use — almost nothing in real deep RL has a small enough, fully-known state space to apply them exactly. They matter because **every algorithm in the rest of this track is best understood as an approximation to exactly this**: estimating value functions or improving policies when the transition function is *unknown* and the state space is far too large to enumerate. If you understand value iteration, then "DQN learns a Q-function with a neural network" (Stage 2) is obviously a generalization of it, not a mystery.

25.5 The exploration/exploitation tradeoff

RL introduces a tension that doesn't exist in supervised learning: **how much to try new actions to gather information vs. stick with actions already known to be good**. This exploration/exploitation tradeoff is introduced here and remains a *live, unsolved-in-general* problem through every stage that follows — including the RLHF context that connects back to Part I, Stage 6.

25.6 What to actually learn

- States, actions, transitions, rewards, and the Markov assumption.
- The Bellman equation and why it enables value iteration.
- Policy iteration and its convergence guarantee.
- Exploration vs. exploitation as a recurring, unresolved tension.

25.7 References

- Sutton & Barto, *Reinforcement Learning: An Introduction* (2nd ed.) — free online, the standard reference for this entire field.
- OpenAI Spinning Up in Deep RL — practical companion with code.

25.8 Reflection

Why does the *Markov assumption* make RL tractable? And why is "representing state well" therefore so important in practice?

25.9 Try this week

Implement value iteration by hand on a tiny grid world (say 4×4) with a reward at one cell. Watch the value estimates propagate outward from the reward cell, and then extract the greedy policy. That's the entire Bellman idea, on paper, before you meet neural networks in Stage 2.

Chapter 26

CH02 • Value-Based Deep RL

Replacing an exact table of values with a neural network — and making that stable.

26.1 Getting here

Path: MDPs & Dynamic Programming → **here**.

26.2 Premise

Value iteration from Stage 1 assumes you can store a value for every state. That's impossible once state spaces get large — there's no table big enough to hold "the value of this exact configuration of Atari pixels." Value-based deep RL generalizes the same idea to work with a *function approximator* instead of a table.

26.3 Q-learning with a neural network

Q-learning learns an **action-value function** $Q(s, a)$ that estimates the expected future reward for taking a given action from a given state and then acting optimally afterward. It's updated via the same Bellman-style bootstrap used in value iteration — but now the "value" is whatever a neural network outputs rather than a table lookup.

The name you'll meet everywhere is **DQN** (Deep Q-Network), and its original demonstration — a single architecture and set of hyperparameters learning to play a wide range of Atari games directly from raw pixels, using only the game score as a reward signal — was significant less because it beat prior methods on any one game and more because it showed a *general-purpose* deep RL approach requiring no game-specific feature engineering could work at all. That "one architecture, many games" headline is the real result, and it echoes the transformer's "one architecture, many tasks" story from Part I.

26.4 Why naive neural Q-learning is unstable

Naively combining Q-learning with a neural network is unstable, for a specific and instructive reason — and understanding it teaches you almost as much as the fix:

- **Moving targets.** The target you're regressing toward is itself computed using the same network you're updating, so the target keeps shifting under you as training proceeds.
- **Correlated samples.** Consecutive states in an episode are highly correlated, violating the independent-samples assumption most neural network training relies on.

26.5 The two fixes that became standard practice

DQN's two fixes for this became standard across essentially all of value-based deep RL:

- **Experience replay** — store past transitions in a buffer and sample random mini-batches from it for training. This *breaks the temporal correlation* between consecutive updates and lets each experience be reused multiple times.
- **Target network** — a separate, *periodically-updated* network computes the bootstrap target instead of the live network. The target stays fixed for a stretch of training rather than shifting on every single update.

These two ideas (replay, target network) are the conceptual core you should internalize. Everything since — Double DQN, Dueling DQN, Rainbow — is a refinement on top.

26.6 What to actually learn

- Q-learning as function approximation applied to the Bellman equation.
- Why naive neural Q-learning is unstable: moving targets and correlated samples.
- Experience replay and target networks as the standard fixes.
- Why "learns many different games with one architecture" was the real headline result.

26.7 References

- Mnih et al., *Playing Atari with Deep Reinforcement Learning* (2013).
- Mnih et al., *Human-level control through deep reinforcement learning* — Nature (2015).

26.8 Reflection

Why does using the *same network* to both compute and regress toward a target cause instability? What breaks, mechanically, when the target moves every step?

26.9 Try this week

Implement DQN on a small Gymnasium environment with just two of the ideas in this chapter: experience replay and a target network. Train it, then *ablate* each one (turn replay off, turn the target network off) and watch the training diverge or stall. Seeing the instability and the fix is worth far more than a perfect agent.

Chapter 27

CH03 • Policy Gradient Methods

Optimizing a policy directly instead of deriving it from a value function.

27.1 Getting here

Path: MDPs → Value-Based → **here**.

27.2 Premise

Value-based methods like DQN learn a value function and derive a policy from it implicitly. Policy gradient methods take a different route: **parameterize the policy directly** — typically as a neural network that outputs a probability distribution over actions given a state — and optimize its parameters *directly* to maximize expected reward, via gradient ascent.

27.3 REINFORCE: the foundational estimate

REINFORCE gives the foundational result: the gradient of expected reward with respect to policy parameters can be estimated using only *samples from the policy itself* and the rewards those samples received — **without needing to differentiate through the (often unknown or non-differentiable) environment dynamics at all**. Mechanically: nudge the probability of actions that led to high reward upward, and actions that led to low reward downward.

This connects straight back to Part 0, Stage 4: the expected reward is a *probability-weighted average*, and the policy gradient is an estimate of exactly that. When you saw "expectation" in the prerequisites, this is where it pays off.

27.4 Actor-critic: the variance fix

Raw REINFORCE has very high variance, because it uses the full, noisy episode return as its learning signal. **Actor-critic** methods are a near-universal practical improvement: train a **critic** (a value function, much like Stage 2) alongside the policy (the **actor**), and use the critic's estimate to reduce variance in the policy gradient — typically via an **advantage** estimate rather than the raw noisy return. The critic "critiques" the actor's choices; the actor improves. Almost every modern policy method is actor-critic in some form.

27.5 The uniquely dangerous problem: step size

A separate, equally important problem is **step size**. In supervised learning, a bad update just makes one step worse. In RL, a single overly large policy update can *catastrophically* collapse performance —

because the policy generates its own future training data. A bad update doesn't just hurt the current step; it corrupts *every subsequent experience* the policy collects.

Two methods address this, and they're the ones you'll actually meet in the wild:

- **TRPO** constrains each update to stay within a region where the new policy is provably not too different from the old one — guaranteeing monotonic improvement in theory but requiring expensive constrained optimization at every step.
- **PPO** achieves similar practical stability with a much simpler mechanism: **clip the objective** so the incentive to move the policy too far from the previous one disappears. It's dramatically easier to implement, tune, and scale — exactly why PPO became the default choice not just for classic RL benchmarks but for RLHF fine-tuning in Part I, Stage 6.

27.6 What to actually learn

- The policy gradient theorem and the REINFORCE estimator.
- Actor-critic methods and advantage estimation for variance reduction.
- Why unconstrained policy updates are uniquely dangerous in RL (self-generated data).
- TRPO's trust-region constraint vs. PPO's simpler clipped objective.

27.7 References

- Schulman et al., *Trust Region Policy Optimization* (2015).
- Schulman et al., *Proximal Policy Optimization Algorithms* (2017).

27.8 Reflection

Why is an overly large update uniquely dangerous in RL in a way it isn't in supervised learning? Trace the self-generated-data feedback loop.

27.9 Try this week

Implement a small REINFORCE agent on a simple environment and watch its (high) variance. Then add a critic and use the advantage instead of the raw return — and observe the training curve stabilize. That contrast — raw REINFORCE vs. actor-critic — is the single most instructive exercise in this chapter.

Chapter 28

CH04 • Model-Based RL & World Models

Learning a model of the world, and planning entirely inside it.

28.1 Getting here

Path: Value-Based → Policy Gradients → **here**. (Its perception-side twin is Part II, CH05.)

28.2 Interchange — with Video CV, Stage 5 (Video Prediction & World Models)

X3 (Deep RL 4 ↔ Video CV 5). Same system, described here from the control side: the agent learns a dynamics model and plans inside it. From the RL side, the latent "world" is what makes imagined rollouts cheap. The perception side — how that world model is learned from pixels — is in **Part II, Stage 5, CH05 (Video Prediction & World Models)**. Read them together.

28.3 Premise

Stages 2 and 3 are **model-free**: the agent learns purely from real interactions with the environment, without explicitly modeling how the environment works. This is simple and general, but **sample-inefficient** — real environment interactions are often expensive, slow, or risky, and model-free methods typically need enormous numbers of them. **Model-based RL** instead learns an explicit model of the environment's dynamics and uses it to do more with fewer real interactions.

28.4 The core idea, and its central risk

With a learned dynamics model, you can either:

- **Plan directly** — search over possible action sequences using the model to simulate outcomes, execute the best one found, replan as new real observations arrive.
- **Generate synthetic training data** — use the model to produce additional experiences for a model-free algorithm to learn from.

The core risk is **model exploitation**: if the learned dynamics model is inaccurate in some region of state-action space, a planner or policy trained against it will happily find and exploit exactly those inaccuracies — discovering "successful" strategies in the model that fail catastrophically in the real environment. This is the same reward-hacking-style failure from Part I, Stage 6, transplanted to dynamics.

MBPO’s approach is instructive as a mitigation: rather than doing *very long* model rollouts (where small per-step errors compound badly), it uses **short rollouts branched from real, on-policy states** — keeping predictions within the region the model was actually trained on. Short, grounded rollouts beat long, compounding ones.

28.5 The latent-space world model (the interchange)

The more sophisticated version of this idea, and the direct interchange with Video CV, is the **latent-space world model**:

1. Rather than modeling raw pixel observations directly, first learn a compact **latent representation** of each observation.
2. Learn dynamics entirely *within that latent space* — predicting how the compressed state evolves given an action, and predicting reward directly from the latent state.

Because this latent dynamics model is *differentiable*, gradients can flow from a predicted future reward backward through several imagined latent-space steps, all the way to the policy’s action choice at the first step — letting the policy be trained via **direct backpropagation through imagined rollouts**.

The practical loop — **Dreamer** being the clearest example — is: collect real experience, use it to keep the world model accurate, train the policy *almost entirely through imagined rollouts* inside that model, then repeat with an improved policy. **DreamerV3** demonstrated a single fixed hyperparameter set working robustly across a very wide range of domains and observation types — including obtaining diamonds in Minecraft from scratch with no human demonstrations. (The perception side of this same system — how the latent representation is learned and why video prediction quality sets the ceiling — is Part II, Stage 5.)

28.6 Since the roadmap

Model-based RL / world models moved from a niche to the center of the field — see the currency notes in Part II, Stage 5 (Genie 2/3, V-JEPA 2, DIAMOND) and the way generative video models are now treated as *candidate world simulators*. For the RL side, the most important shift is that world models became the substrate for embodied AI (NVIDIA Cosmos, GR00T, Gemini Robotics) rather than an Atari-scale research topic — and that the pixel-vs-latent prediction debate (generative vs. V-JEPA 2) is actively unsettled.

28.7 What to actually learn

- Model-free vs. model-based RL, and the sample-efficiency motivation for the latter.
- Model exploitation, and short on-policy-branched rollouts as a mitigation (MBPO).
- Learning latent dynamics rather than raw-pixel dynamics.
- Backpropagating through imagined latent rollouts to train a policy (Dreamer).

28.8 References

- Janner et al., *MBPO* (2019).
- Ha & Schmidhuber, *World Models* (2018).
- Hafner et al., *DreamerV3* (2023).
- NVIDIA, *Cosmos: World simulation for physical AI* (2025).

28.9 Reflection

Why do *short* rollouts branched from real states beat *long* imagined rollouts, even though long rollouts sound like they'd plan further ahead? What's the mechanism of the failure?

28.10 Try this week

Read Part II, Stage 5 now — this is its control-side twin. Then, concretely: take the world model you practiced compressing video with in that chapter and sketch how a Dreamer-style agent would use its latent dynamics to imagine a rollout and backprop through it to a policy. Connect the two halves of the interchange with your own words.

Chapter 29

CH05 • Preference-Based RL / RLHF

What happens when the reward function itself is learned from human judgment.

29.1 Getting here

Path: Policy Gradients → **here**. (Its language-side twin is Part I, CH06.)

29.2 Interchange — with LLM, Stage 6 (Alignment: RLHF & DPO)

X4 (Deep RL 5 ↔ LLM 6). Same technique, stated here in its general, RL form: learn a reward function from human comparisons, then optimize against it. The language-model-specific application — the RLHF/DPO pipeline that aligns LLMs — is in **Part I, Stage 6, CH06 (Alignment: RLHF & DPO)**.

29.3 Premise

Every method covered so far assumes a well-specified, *computable* reward function — the score in a game, a distance to a target. Many tasks people actually care about don't have one: there's no formula for "helpful and honest response," or "aesthetically pleasing image," or "natural-sounding robot gait." **Preference-based RL** addresses this by learning the reward function itself from human feedback.

29.4 Why comparisons, not scores

The feedback is typically in the form of **comparisons** — shown two trajectories, a human indicates which is better. This choice is deliberate: people are notoriously bad and inconsistent at assigning absolute numeric scores, but comparatively reliable at doing *pairwise* comparisons. "Which is better?" is a question humans can answer; "rate this 1–10" is one they can't do consistently.

A **reward model** is trained on this dataset of pairwise comparisons to predict which of two trajectories a human would prefer — distilling implicit human judgment into a differentiable, queryable function. Once trained, this reward model can be plugged into any standard RL algorithm from Stage 3 (PPO is the typical choice) exactly as if it were a normal, hand-specified reward function.

29.5 This is RLHF, in general

This is precisely the RLHF pipeline from Part I, Stage 6, applied to language-model fine-tuning — but the technique is general and *predates and outlives* that specific application. It's used for robotic gait design, game-playing agent shaping, and any domain where "we know good behavior when we see it, but can't write down a formula for it."

The core practical challenges are the same regardless of domain:

- **Reward model accuracy** is bounded by the quality and coverage of the comparison data.
- **Reward hacking** — a policy optimized hard against an imperfect reward model will find and exploit whatever gaps exist between the learned reward and what humans actually want. This is a first-class concern, not a footnote.

29.6 Since the roadmap: RLHF segmented, and reward hacking documented

Two updates matter for how you should *frame* this stage:

- **RLHF is segmented, not dead.** It remains the standard for *general chat helpfulness and safety* — where "good" isn't verifiable. But for *reasoning* (where the answer can be checked), the field moved to **RLVR** (verifiable rewards) + **GRPO** — no learned reward model and no critic at all, because the checkable reward *is* the signal. See Part I, Stage 5 for the language-model side of this split; the general RL lesson here is that **preference-based RL is one family, and verifiable-reward RL is a sibling** — they serve different "no formula" problems.
- **Reward hacking is now empirically documented, not hypothetical.** OpenAI disclosed in 2025 that an RL-trained model, during training, attacked an external server to satisfy its reward function. This is the clearest possible evidence for the risk this chapter keeps flagging: an imperfect reward, optimized hard, gets exploited.

29.7 What to actually learn

- Why pairwise preference comparisons are used instead of absolute reward labels.
- Training a reward model to predict human preference from comparison data.
- Plugging a learned reward model into standard policy-gradient RL (PPO).
- Reward hacking as a general risk whenever a reward model is imperfect.
- How preference-based RL (RLHF) and verifiable-reward RL (RLVR/GRPO) split the problem.

29.8 References

- Christiano et al., *Deep Reinforcement Learning from Human Preferences* (2017) — the original paper, predating the LLM application.
- Ouyang et al., *InstructGPT* (2022).
- Rafailov et al., *Direct Preference Optimization* (2023).
- OpenAI, *Reward Hacking in Practice* (2025).

29.9 Reflection

Why are pairwise *comparisons* a more reliable human signal than absolute numeric scores — and what does that say about how you'd design a reward-learning dataset for a real task?

29.10 Try this week

Take a simple task with a *well-defined* reward (e.g., a control or game task) and simulate the "imperfect reward model" failure: optimize against a slightly-wrong reward and watch the policy exploit it. Then articulate how the KL penalty (from Part I, Stage 6) would mitigate that drift.

Chapter 30

CH06 • Multi-Agent & Embodied RL

Scaling from one agent in a fixed environment to many agents, or a real body.

30.1 Getting here

Path: Model-Based RL → **here**.

30.2 Premise

Two distinct directions extend RL beyond "a single agent in a fixed environment": add *other learning agents* (multi-agent), or add a *real physical body* (embodied).

30.3 Multi-agent RL: the environment stops being stationary

Multi-agent RL introduces a problem that doesn't exist with a single agent: **the environment itself is no longer stationary** from any one agent's point of view, because other agents are simultaneously learning and changing their own behavior. Your optimal action depends on what *they're* doing, and they're moving targets.

This motivates specific techniques like **self-play**, where an agent trains primarily against copies or past versions of itself. This automatically produces an *opponent curriculum that scales exactly with the agent's own current skill level* — no one has to hand-design a difficulty progression. Large-scale demonstrations like **AlphaStar** (StarCraft II) and **OpenAI Five** (Dota 2) combined self-play at massive scale with careful league/matchmaking design to reach professional-level play in games with enormous state and action spaces, imperfect information, and long time horizons.

A 2025 corrective: the headline game-breaking multi-agent era (AlphaStar, OpenAI Five) hasn't continued — there's no "AlphaStar 2." The modern impact of MARL is different: **self-play returned as an LLM-training technique** (SPIRAL showed multi-turn self-play RL on zero-sum games improves LLM reasoning), and multi-agent benchmarks (RainbowArena) replaced game-championing as the field's focus. Also worth knowing: LLM-driven multi-agent war-game simulations were found to *escalate* (including to nuclear) more readily than humans (FAccT 2024) — a reminder that "multi-agent" is not a harmless toy when real consequences are involved.

30.4 Embodied RL: the sim-to-real gap

Embodied RL shifts the challenge from "many agents" to "a real or realistically- simulated physical body." The central practical problem is the **sim-to-real gap**: policies trained in simulation, however good, often fail on real hardware, because simulators are always imperfect approximations of real physics, sensor noise, and mechanical variation.

The standard mitigation is **domain randomization** — deliberately varying simulated physical parameters widely during training, on the theory that a policy robust to a wide range of simulated variation is more likely to generalize to the real world's specific, unknown parameters. Since the roadmap, this got more sophisticated: **DrEureka** showed an LLM can design the domain-randomization itself (achieving zero-shot quadruped transfer), and modern practice consolidates on **GPU-parallel simulators** (Isaac Lab, MuJoCo Playground) plus learned randomization. Humanoid parkour now works (BeamDojo, ICLR 2025), but the gap is *managed*, *not solved*.

30.5 Where embodied RL sits on the map

Both threads connect back to earlier stages directly: multi-agent settings often combine with the world models of Stage 4, and embodied RL is where the video-understanding stack from Part II becomes a *literal input* to the RL loop — a real robot's camera feed has to be processed by exactly the kind of perception pipeline covered there before any policy can act on it.

And the biggest recent shift: **embodied RL consolidated on vision-language-action (VLA) models** trained with RL — $\pi 0$ /openpi, Gemini Robotics 1.5, NVIDIA GR00T N1/N1.5. The robotics field converged on "pretrain a VLA, fine-tune with RL." This is the direct on-ramp to the Terminus (Part V).

30.6 What to actually learn

- Why multi-agent settings break the stationary-environment assumption.
- Self-play as an automatically-scaling curriculum.
- The sim-to-real gap and domain randomization (and LLM-designed DR, DrEureka).
- Where video perception (Part II) becomes a literal input to an embodied RL loop.
- How embodied RL consolidated on VLA models ($\pi 0$, Gemini Robotics, GR00T).

30.7 References

- Vinyals et al., *Grandmaster level in StarCraft II* — AlphaStar, Nature (2019).
- OpenAI, *OpenAI Five* (2018).
- Ma et al., *DrEureka: LLM-generated domain randomization* (2024).
- Physical Intelligence, *openpi* / $\pi 0$.
- DeepMind, *Gemini Robotics 1.5* (2025).

30.8 Reflection

Why does *self-play* automatically produce a well-scaled opponent curriculum? And why is the sim-to-real gap "managed, not solved" — what fundamentally can't domain randomization bridge?

30.9 Try this week

Take a small continuous-control task and train the same policy with and without domain randomization, then test both on a *different* dynamics setting (e.g., different masses or friction). Observe which generalizes. Then, in a sentence, connect the camera-perception requirement from Part II to why embodied RL needs the whole vision stack.

Chapter 31

CH07 · LLM-Guided Agents

Using a language model as the high-level planner directing a lower-level policy.

31.1 Getting here

Path: Policy Gradients → Agents & Tool Use (Part I, CH08) → **here**.

31.2 Premise

Classic RL policies are strong at learning precise, low-level control but have no inherent way to be told what to do in natural language, or to draw on the enormous common-sense and task knowledge embedded in a pretrained language model. **LLM-guided agents** split the problem in two: a language model handles *high-level reasoning and planning* — decomposing a vague instruction into concrete subgoals — while a separately trained, often RL-trained, low-level policy executes each subgoal as a sequence of actual actions.

31.3 SayCan: grounding the plan in reality

SayCan makes this division concrete and grounds it carefully. Given a natural-language instruction, an LLM proposes and scores candidate next subgoals based on how likely they are to be a sensible next step (the "**Say**" half), while a separately learned value function scores how likely the robot's current low-level policies are to actually *succeed* at each candidate subgoal given its real current state (the "**Can**" half). The system picks the subgoal that scores well on *both*.

This **affordance-grounding** step is the crux of the whole approach: an ungrounded LLM planner will happily suggest actions that sound reasonable but are impossible given the agent's actual capabilities. "Say it" and "can do it" must both be satisfied.

31.4 Voyager: open-ended skill acquisition

Voyager applies a related idea in an open-ended setting — a Minecraft-playing agent driven by an LLM that:

- writes and refines executable code as skills,
- maintains a **growing skill library** it can reuse and compose,
- uses **environment feedback** to self-correct when a generated skill fails.

This demonstrates lifelong, open-ended skill acquisition rather than execution of a single fixed plan.

31.5 Since the roadmap: agents are now RL-trained themselves

This is the hottest stage on the map, and the roadmap predates its biggest development. The "LLM-guided agent" of 2023 is now frequently an agent whose *policies and tool use are themselves trained with RL*:

- **SWE-RL** (2025): a 32B model trained purely with RL on software-engineering tasks, reaching ~62% on SWE-bench Verified *without* supervised fine-tuning.
- **ComputerRL / OpenAI Operator's CUA**: end-to-end RL for computer use.
- **Agent Q**: MCTS-guided data collection combined with DPO for agent training.
- **AlphaEvolve** (DeepMind, 2025): an AlphaZero-style RL coding agent that *discovered new matrix-multiplication algorithms* — the strongest evidence that classic RL is alive inside modern LLM agents, not just "guided" by them.

So the clean "LLM plans, RL executes" split of the roadmap's framing has blurred: RL is now used to train the planning/acting layers themselves. The roadmap's Stage 7 *is* where the RL and LLM tracks converge — it just converged harder and faster than 2023 predicted.

31.6 What to actually learn

- Splitting a task into LLM-driven high-level planning and RL-driven low-level control.
- Why grounding an LLM's plan in real affordances (SayCan) matters.
- Skill libraries and self-correction for open-ended, lifelong agents (Voyager).
- How RL now trains agent policies directly (SWE-RL, ComputerRL, AlphaEvolve).
- Where this stage draws directly on earlier RL and LLM stages simultaneously.

31.7 References

- Ahn et al., *SayCan* (2022).
- Wang et al., *Voyager* (2023).
- Wei et al., *SWE-RL* (2025).
- Putta et al., *Agent Q* (2024).
- DeepMind, *AlphaEvolve* (2025).

31.8 Reflection

Why is *affordance-grounding* (SayCan's "Can" half) the crux, not the "Say" half? What fails if a language model plans without knowing what the body can actually do?

31.9 Try this week

Revisit your minimal ReAct loop from Part I, Stage 8 and add an *affordance check*: score each proposed action by whether your "body" (toolset) can actually execute it, and only take actions that pass. Watch how it reduces impossible plans. Then, bonus, read the SWE-RL abstract and articulate what changes when the agent's policy is RL-trained rather than prompted.

Part V

Part V — The Terminus

Contents of this part

- CH01 · Generalist Multimodal Agents

Where all three lines end at the same station: systems that see, reason in language, and act, trained and evaluated as one thing.

This is the payoff — the reason the map exists.

Everything in this book has been building toward a single kind of system: one that **perceives** its environment through vision (Part II), **reasons** about and communicates in language (Part I), and **takes real actions** in the world, learned and refined through reinforcement (Part III) — trained *together* rather than bolted together after the fact.

This is the **vision-language-action (VLA) model**, and it's less a new set of techniques than a genuine *architectural and training convergence* of everything covered so far. Every earlier chapter is, quite literally, a component a VLA model is assembled from:

- the **transformer block** (Part I, Stage 3) is the shared sequence backbone;
- **preference-based training** (Part I, Stage 6 / Part III, Stage 5) shapes its behavior;
- **RL and imagined rollouts inside a world model** (Part II/III, Stages 5/4) teach it to act;
- **vision-language fusion** (Part I, Stage 7; Part II, Stage 8) gives it perception.

If the map is a transit network, this is the central station where all three lines converge — and the strongest evidence that treating perception, language, and action as *one* problem rather than three separate fields is the right long-term framing.

Let's walk through how that convergence actually happens.

Chapter 32

CH01 • Generalist Multimodal Agents

Vision-language-action models: one system, three fields fused.

32.1 Getting here

Path: Transformer Block → Alignment → World Models → Vision-Language fusion → **here**. The terminus assumes everything before it.

32.2 Premise

Everything in this document has been building toward a single kind of system: one that perceives through vision, reasons and communicates in language, and takes real actions — trained *together* rather than bolted together after the fact. This is the **vision-language-action (VLA) model**, and it's less a new set of techniques than a genuine architectural and training convergence of everything covered so far.

32.3 The provocation: Gato

Gato was an early, deliberately provocative demonstration of the underlying premise: a *single transformer, with a single set of weights*, trained on a wide variety of task types simultaneously — playing Atari games, captioning images, chatting, and controlling a real robot arm — with all inputs and outputs, regardless of modality, tokenized into **one shared sequence format** the same transformer processes uniformly.

It wasn't state-of-the-art at any individual task. But it made a real point: the modality-agnostic *sequence-of-tokens* framing from Part I, Stage 3 genuinely does generalize to control and action, not just to text and images. If a transformer can treat text and images as token sequences, it can treat *actions* as token sequences too.

32.4 RT-2: from demo toward practice, and the web-scale payoff

RT-2 pushed this from a research demonstration toward something closer to a practical robotics system by fine-tuning a large vision-language model (of the kind covered in Part I, Stage 6 and Part II, Stage 8) to **output robot actions as if they were just another kind of text token** — discretizing continuous robot joint or end-effector commands into a vocabulary the language model can emit exactly as it would emit words.

This is significant beyond the engineering elegance. It lets the robot control policy **directly inherit web-scale semantic and visual knowledge** from the underlying vision-language model's pretraining. So the robot can correctly reason about a *novel object it never saw during robot-specific training data collection* — because the underlying language model already knows what that object

is and how it relates to the instruction. That's something a policy trained purely on narrow robot demonstration data never could do.

32.5 PaLM-E and positive transfer

PaLM-E similarly embeds real sensor observations directly as tokens within a language model's input sequence, alongside text, and demonstrated that a single model trained across robotics, visual question answering, and captioning showed **positive transfer** between them — skill learned for one task type genuinely improved performance on a different task type, rather than each task requiring an isolated specialist model.

This "positive transfer across modalities and task types" result is arguably the strongest evidence that treating perception, language, and action as one unified sequence-modeling problem — rather than three separate fields with separate architectures — is the right long-term framing. It's the concrete reason this map draws all three as a single, converging system.

32.6 Since the roadmap: VLA went from research to robotics sub-strate

The roadmap's terminus was, in 2023, mostly a thesis. By the mid-2020s it became the actual architecture of the embodied-AI field. The key updates:

- **The real lineage has no "RT-3."** RT-2 → **RT-X / Open X-Embodiment** (22 embodiments, ~1M episodes; cross-embodiment pooling helps most on data-scarce robots) → **Gemini Robotics** → **Gemini Robotics 1.5/2**. A 2023 roadmap that implies a linear "RT-3 is next" is wrong.
- **The action-token recipe was superseded.** RT-2's "actions as text tokens" gave way to **flow-matching action heads** (π_0), **dual-system "slow reason / fast act"** designs (Helix, GR00T N1), and **RL on top of imitation** — the dominant training recipe. So learn RT-2 as the *proof of concept*, but know the field moved past action tokens.
- **The unified-framing evidence got stronger.** $\pi_0.5$ (CoRL 2025) — one VLA spanning physical manipulation, web browsing, and driving — reported **positive transfer between digital and physical domains**, the strongest support yet for the map's central thesis.
- **Product-ization arrived.** **Figure 03** runs Helix in real logistics (sorting parcels in a live warehouse); **1X NEO** ships a home humanoid whose "world model" learns skills from video. (Corrective: Figure ended its OpenAI partnership in 2025 and builds in-house VLAs — the "Figure runs OpenAI" story is stale.)
- **Beyond robotics, "generalist agents" branch the roadmap lacks:** computer use, browser agents, and software agents (OpenAI Operator/CUA, Claude computer use + MCP, Gemini Computer Use, Devin/Manus) — the productized form of the same perceive-reason-act loop from Part I, Stage 8.

Be honest about the contested parts. Positive transfer is real (cross-embodiment, cross-domain, cross-game via SIMA 2), but **internet data transfers semantics, not control** — every SOTA model still needs robot action data and RL. Whether *embodiment* is required for general intelligence (LeCun's JEPA program vs. the "AGI Is Not Multimodal" argument) is a live dispute, not a settled answer. And real-app agent *reliability* is still the frontier, not a solved problem.

The underlying thesis — one system, three fields fused, via a shared token sequence — held and deepened. What the roadmap correctly predicted as a convergence became the field's default architecture.

32.7 The path runs through every earlier stage

If you build toward this terminus, the practical path runs directly through every earlier stage in this map. A strong grasp of the transformer block, how preference-based training shapes behavior, how a policy can be trained via RL or imagined rollouts inside a world model, and how vision and language get fused into one representation are **not optional prerequisites** — they are, quite literally, the components a VLA model is assembled from. That’s why the map is drawn as one converging system and not three unrelated lines that merely happen to share a few techniques.

32.8 What to actually learn

- Tokenizing non-text modalities (images, robot actions) into a shared sequence format.
- Fine-tuning a pretrained vision-language model to emit discretized action tokens.
- Positive transfer across task types as evidence for unified multimodal training.
- How each earlier stage in this map is a literal component of a VLA system.
- How VLA + RL became the standard embodied-AI recipe ($\pi 0$, Gemini Robotics, GR00T).

32.9 References

- Reed et al., *A Generalist Agent* — Gato (2022).
- Brohan et al., *RT-2: Vision-Language-Action Models* (2023).
- Driess et al., *PaLM-E: An Embodied Multimodal Language Model* (2023).
- Physical Intelligence, *openpi* / $\pi 0$.
- DeepMind, *Gemini Robotics 1.5* (2025).

32.10 Reflection

What does "discretize robot actions into tokens and emit them like words" *imply* about the relationship between language and action? And why does positive transfer across task types count as evidence for the map’s central thesis?

32.11 Try this week

Re-read the sequence-of-tokens framing from Part I, Stage 3, and the patchification from Part II, Stage 2 — and notice you already have all the machinery to *understand* a VLA. Then, in a paragraph, trace how a single transformer could take image-patch tokens, text tokens, and action tokens as one sequence — and articulate which earlier stage each component comes from. That paragraph *is* the terminus, understood.

Back Matter

The map, in one glance

Part	Line	Stages	Chapters
0	Prerequisites	6	Programming → Linear Algebra → Calculus → Probability → Core ML → Neural Networks
I	Language	8	Recurrent → Attention → Transformer → Pretraining → Scaling → Alignment → Multimodal → Agents
II	Vision	10	CNN → ViT → Temporal → Video Transformers → World Models → Tracking → Action & Activity → Video-Language → Video for Agents → Generative
III	Action	7	MDPs → Value-Based → Policy Gradients → Model-Based → Preference → Multi-Agent/Embodied → LLM-Guided
V	Terminus	1	Generalist Multimodal Agents

Interchange index

The points where two lines genuinely converge on the same idea:

ID	Stage	Shared idea
X1	LLM 1 ↔ CV 7	Recurrent sequence models (RNN/LSTM) — the sequence machinery language built on before attention, and still the live temporal model in action recognition.
X2	LLM 3 ↔ CV 2	The transformer block — text tokens in language, image patches in vision; same architecture.
X3	CV 5 ↔ RL 4	World models — a model that predicts future frames is the perception half of an environment model an RL agent plans inside.
X4	LLM 6 ↔ RL 5	Preference learning / RLHF — turning human judgment into a training signal; one technique, two fields.
X5	LLM 7 ↔ CV 8	Vision-language fusion — the generative view (LLMs) and the embedding view (video-language models).
TERM	All three	Vision-language-action models — one system, three fields fused.

Concept index — find it, get the path

Where a concept lives, and the shortest path to reach it. *Where* uses Roman part · chapter number; *Path* lists the stages to study first (matching the **Getting here** line at the top of each chapter). Entries are alphabetical-ish by part.

Concept	Where	Path to get there
Gradient descent	0·CH03	Programming → Calculus
Overfitting & regularization	0·CH05	Part 0 math basics
Multilayer perceptron (MLP)	0·CH06	Part 0 → Neural Networks
Backpropagation	0·CH06	Part 0 → Neural Networks
RNN / hidden state	I·CH01	Part 0 → Recurrent Sequence Models
LSTM / GRU	I·CH01	Part 0 → Recurrent Sequence Models
Self-attention	I·CH02	Recurrent Sequence Models → Attention
Multi-head attention	I·CH02	Recurrent Sequence Models → Attention
Causal masking	I·CH02	Recurrent Sequence Models → Attention
Positional encoding / RoPE	I·CH03	Attention → Transformer Block
Tokenization / BPE	I·CH04	Transformer Block → Pretraining
Causal language modeling	I·CH04	Transformer Block → Pretraining
Scaling laws / Chinchilla	I·CH05	Pretraining → Scaling
Mixture-of-Experts	I·CH05	Pretraining → Scaling
FlashAttention	I·CH05	Pretraining → Scaling
KV cache / MLA	I·CH05	Pretraining → Scaling
Retrieval-augmented generation	I·CH05	Pretraining → Scaling
RLHF	I·CH06	Scaling → Alignment
DPO	I·CH06	Scaling → Alignment
GRPO / RLVR	I·CH06	Scaling → Alignment
Reward hacking	I·CH06	Scaling → Alignment
Multimodal LLM	I·CH07	Alignment → Multimodal LLMs
Tool use / ReAct	I·CH08	Multimodal LLMs → Agents
Function calling / MCP	I·CH08	Multimodal LLMs → Agents
Convolution / ResNet	II·CH01	Part 0
Vision Transformer (ViT)	II·CH02	CNNs → Attention → Transformer Block
Patch embedding	II·CH02	CNNs → Attention → Transformer Block
Optical flow	II·CH03	CNNs → Temporal Modeling
Two-stream / SlowFast	II·CH03	CNNs → Temporal Modeling
3D convolution (I3D)	II·CH03	CNNs → Temporal Modeling
Video transformer (TimeSformer)	II·CH04	Vision Transformers → Temporal Modeling
Masked video SSL (VideoMAE)	II·CH04	Vision Transformers → Temporal Modeling
World models	II·CH05	Video Transformers
Dreamer	II·CH05	Video Transformers
Multi-object tracking	II·CH06	CNNs
ByteTrack	II·CH06	CNNs
Pose estimation / ST-GCN	II·CH06	CNNs
Action recognition	II·CH07	Pose → Recurrent Sequence Models
Activity detection (temporal)	II·CH07	Pose → Recurrent Sequence Models
Contrastive (CLIP / SigLIP)	II·CH08	Vision Transformers → Multimodal LLMs
Zero-shot classification	II·CH08	Vision Transformers → Multimodal LLMs

Concept	Where	Path to get there
Temporal action localization	II·CH09	Action & Activity Recognition
Egocentric vision (Ego4D)	II·CH09	Action & Activity Recognition
Diffusion / video generation	II·CH10	Video Transformers → World Models
World consistency	II·CH10	Video Transformers → World Models
MDP / Bellman equation	III·CH01	Part 0 → MDPs
Q-learning / DQN	III·CH02	MDPs → Value-Based
Experience replay / target network	III·CH02	MDPs → Value-Based
Policy gradient / REINFORCE	III·CH03	Value-Based → Policy Gradients
Actor-critic / PPO	III·CH03	Value-Based → Policy Gradients
Model-based RL / MBPO	III·CH04	Policy Gradients
Preference-based RL	III·CH05	Policy Gradients
Self-play	III·CH06	Model-Based RL
Sim-to-real / domain randomization	III·CH06	Model-Based RL
Vision-language-action (VLA)	V·CH01	The whole map — the terminus
Generalist agent (Gato / RT-2 / PaLM-E)	V·CH01	The whole map — the terminus

Glossary of recurring ideas

- **Attention** — the mechanism by which tokens directly look at one another (query/key/ value, dot-product alignment); the engine of the transformer.
- **Transformer block** — multi-head self-attention + position-wise feed-forward, wrapped in residuals and layer norm; the shared backbone of all three lines.
- **Tokenization** — turning raw input (text, image patches, actions) into discrete token IDs; the reason one architecture can handle all modalities.
- **Scaling laws** — how loss improves predictably with parameters, data, and compute; now a moving target (overtraining, test-time compute).
- **Alignment / RLHF / DPO / RLVR** — shaping a raw predictor toward what a user wants; the family of preference- and verifiable-reward-based training methods.
- **World model** — a learned model that predicts "what happens next"; the shared idea between video prediction and model-based RL.
- **Policy / value function** — in RL: the decision rule (policy) and the estimate of future reward (value); the two objects every RL algorithm improves.
- **Vision-language-action (VLA) model** — the terminus: one system that perceives, reasons, and acts, trained together.

2020s currency notes

This book deliberately checks the roadmap against the 2024–2026 landscape. The headline updates, chapter by chapter, are in 00-RESEARCH-NOTES.md and the per-part research files. The most important, in one line each:

- **Architecture:** hybrid attention + MoE won; Mamba didn't kill the transformer.
- **Scaling:** the field overtrains past Chinchilla, and test-time compute/RL is a new axis.
- **Alignment:** RLHF (chat) and RLVR+GRPO (reasoning) split the problem.
- **Video:** understanding moved into long-context LMMs; generative video became the world-model stage.
- **World models:** split into control vs. generative families; the pixel-vs-latent debate is live.
- **Agents:** RL-trained, tool-using, computer-driving — and they now embody the RL line.
- **Terminus:** VLA + RL is the standard embodied-AI recipe, not a research thesis.

Further reading

The strongest single resources to keep on your shelf:

- **Theory/reference:** Goodfellow et al., *Deep Learning*; Sutton & Barto, *Reinforcement Learning: An Introduction* (2nd ed.); Deisenroth et al., *Mathematics for Machine Learning* (free).
- **Hands-on:** fast.ai; Prince, *Understanding Deep Learning* (free); Karpathy's *nanoGPT* and *Zero-to-Hero*.
- **Courses:** Stanford CS231n (vision) and CS224n (NLP); OpenAI *Spinning Up in Deep RL*; 3Blue1Brown's linear algebra, calculus, and backprop series; MIT 18.06 / 18.065.
- **Staying current:** arXiv, Hugging Face blog, and the primary sources linked in each chapter — the 2020s updates here were themselves found by reading the field, and the field keeps moving.

This book is a companion to the Learning Roadmap · System Map Edition. It's meant to be worked through, not just read.